NGCDF Allocation Analysis: Rift Valley (Group A) and Central (Group B) Counties (2014-2030)

Stephen Nzambu Ndundu (SDS6/49567/2025)
Susan Leah Wangari (SDS6/49245/2025)

2026-02-12

Table of contents

List of Figures

List of Tables

NGCDF ALLOCATION ANALYSIS: RIFT VALLEY (GROUP A) AND CENTRAL (GROUP B) COUNTIES (2014–2030)

GROUP 09

12 February 2026

NAIROBI UNIVERSITY

FACULTY OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF MATHEMATICS

NAMES OF CONTRIBUTORS:

1) STEPHEN NZAMBU NDUNDU
2) SUSAN LEAH WANGARI

REG NUMBERS:

1) SDS6/49567/2025
2) SDS6/49245/2025

CONTRIBUTION: 50:50

DECLARATION

We, the undersigned, declare that this research report titled "NGCDF Allocation Analysis: Rift Valley (Group A) and Central (Group B) Counties (2014–2030)" is our original work and has not been submitted for any academic award in any other university or institution of higher learning.

Stephen Nzambu Ndundu SDS6/49567/2025

Signature: ______________________________ Date: ____________________

Susan Leah Wangari SDS6/49245/2025

Signature: ______________________________ Date: ____________________

Supervised by;

____

Department of Mathematics, Faculty of Science and Technology

Signature: ______________________________ Date: ____________________

DEDICATION

This work is dedicated to our families for their unwavering support, encouragement, and patience throughout the period of this research. Their sacrifices and understanding have been instrumental in the successful completion of this study.

We also dedicate this work to the people of Trans Nzoia, Nandi, Uasin Gishu, Murang'a, Kiambu, and Kirinyaga counties, whose development needs inspired this analysis of public resource allocation.

ACKNOWLEDGEMENT

We wish to express our sincere gratitude to Almighty God for His grace, wisdom, and strength throughout this research journey.

Our profound appreciation goes to our supervisor in the Department of Mathematics, Faculty of Science and Technology, for invaluable guidance, constructive criticism, and continuous support throughout this study. Their expertise and mentorship have been instrumental in shaping this research.

We are grateful to the National Government Constituencies Development Fund (NGCDF) Board for providing the allocation data that formed the foundation of this analysis. Special thanks to the Kenya National Bureau of Statistics (KNBS) for the Consumer Price Index data that enabled our inflation-adjusted analysis.

We acknowledge the Independent Electoral and Boundaries Commission (IEBC) for the registered voter data that facilitated accurate per-voter and constituency-level analysis.

Our gratitude extends to our colleagues and classmates who provided moral support and constructive feedback during the development of this report.

Finally, we thank all those who directly or indirectly contributed to the successful completion of this research. May God bless you all.

ABBREVIATIONS AND ACRONYMS

ANOVA - Analysis of Variance CAGR - Compound Annual Growth Rate CDF - Constituencies Development Fund CDFC - Constituency Development Fund Committee CPI - Consumer Price Index CRA - Commission on Revenue Allocation CV - Coefficient of Variation GIS - Geographic Information System IEBC - Independent Electoral and Boundaries Commission KES - Kenya Shillings KNBS - Kenya National Bureau of Statistics MP - Member of Parliament NGCDF - National Government Constituencies Development Fund OLS - Ordinary Least Squares SDG - Sustainable Development Goal USAID - United States Agency for International Development WB - World Bank

ABSTRACT

This study analyzed the allocation patterns of the National Government Constituencies Development Fund (NGCDF) across six Kenyan counties, grouped into two regional clusters: Group A, Trans Nzoia, Nandi, and Uasin Gishu (Rift Valley), and Group B — Murang'a, Kiambu, and Kirinyaga (Central), over the period 2014 to 2026, with linear projections through 2030. The research was guided by four objectives: to document allocation totals and year-on-year trends, to analyze administrative and constituency-level distribution patterns, to assess intra-county equity and inflation-adjusted (real) allocation trends, and to provide a comparative, data-driven assessment across the two regional groups. The study employed a quantitative research design utilizing secondary data obtained from the NGCDF Board, the Independent Electoral and Boundaries Commission (IEBC), and the Kenya National Bureau of Statistics (KNBS) Consumer Price Index series. Data analysis was conducted using the R programming language, employing descriptive statistics, trend and forecasting analysis (OLS regression), Gini coefficients, coefficients of variation, and inferential tests (ANOVA, Kruskal-Wallis, paired t-tests, and Pearson correlation). The findings revealed consistent nominal growth in NGCDF allocations across all six counties between 2014 and 2026, though real (CPI-adjusted) growth was substantially more modest once Kenya's cumulative inflation of approximately 77% over the period is accounted for. Kiambu recorded the highest absolute allocations within Group B, driven largely by its twelve constituencies, while intra-county equity, measured through Gini coefficients and coefficients of variation, varied meaningfully both across counties and over time. Per-voter allocation analysis offered a more equitable lens than absolute totals, revealing that constituencies with smaller voter rolls sometimes received disproportionately higher funding per voter. National ranking analysis situated all six counties within the broader landscape of Kenya's 47 counties, while linear forecasts to 2030 projected continued nominal growth contingent on national fiscal conditions. The study contributes a replicable analytical framework for monitoring NGCDF allocations across regionally distinct county clusters and offers recommendations for enhancing the transparency, equity, and monitoring of constituency-level development financing in Kenya.

Keywords: NGCDF, constituency development, resource allocation, per-voter analysis, Gini coefficient, fiscal decentralization, Kenya, Rift Valley, Central Kenya

CHAPTER ONE: INTRODUCTION

1.1 Background of the Study

The National Government Constituencies Development Fund (NGCDF) represents one of Kenya's most significant decentralized development initiatives, designed to address regional imbalances and promote equitable distribution of national resources. The fund channels resources directly to the constituency level, empowering local communities to identify, prioritize, and implement development projects that address their specific needs.

The concept of constituency-level development funding emerged from the recognition that centralized resource allocation often failed to address the unique development challenges facing different regions of the country. The NGCDF operates on the principle of participatory development, where constituency-level committees, comprising local leaders, technical experts, and community representatives, identify and prioritize projects for funding, aligning with global trends toward decentralization and community-driven development.

1.1.1 Historical Development of NGCDF

The fund was established under the National Government Constituencies Development Fund Act (No. 30 of 2015, Cap. 414A of the Laws of Kenya) (Republic of Kenya, 2015) to channel a portion of national government revenues directly to parliamentary constituencies for grassroots-level development. Its core mandate is to reduce regional development disparities by ensuring that even the most remote and underserved constituencies receive a predictable annual allocation for infrastructure, education, health, and community projects.

The fund has undergone several legislative refinements since its inception, most notably its renaming and restructuring to align with the devolved system of government established by the 2010 Constitution. These changes introduced enhanced governance structures, including the NGCDF Board, constituency-level committees, and strengthened accountability mechanisms. From an initial focus on education infrastructure, the fund has expanded to support a wide range of development interventions, including healthcare facilities, water projects, roads, and economic empowerment initiatives.

1.1.2 NGCDF Allocation Framework

Each of Kenya's 290 parliamentary constituencies receives an annual allocation, the size of which is primarily determined by a national revenue-sharing formula. Allocations are administered through Constituency Development Fund Committees (CDFCs) chaired by the local Member of Parliament, with oversight from the NGCDF Board. The allocation framework comprises several components:

Equal Share Component: Each constituency receives a base allocation to ensure minimum funding levels regardless of population or other factors.

Population-Based Component: A significant portion of the fund is allocated based on constituency population, recognizing that areas with larger populations require more resources to address development needs.

Poverty Index Component: Constituencies with higher poverty levels receive additional allocations to address historical disadvantages and promote equity.

Emergency Reserve: A portion of the fund is maintained as an emergency reserve to respond to unforeseen circumstances such as natural disasters or emergencies requiring immediate intervention.

Because counties with more constituencies naturally receive a larger county-level total under the equal-share component, cross-county comparisons of absolute totals can be misleading unless read alongside per-constituency and per-voter measures, a consideration that motivates much of the comparative analysis in Chapter Four of this report.

1.1.3 Overview of the Study Counties

This report analyses six counties grouped by geographic region into two clusters of three counties each.

Group A: Rift Valley Region encompasses Trans Nzoia, Nandi, and Uasin Gishu. These are predominantly agricultural counties in the former Rift Valley Province, sharing borders and, to some extent, similar economic profiles anchored in maize farming, tea, and dairy. Eldoret, the regional hub city in Uasin Gishu, gives this cluster a notable urban centre that influences economic activity across the broader region.

Group B: Central Region comprises Murang'a, Kiambu, and Kirinyaga. These counties form part of the Mount Kenya foothill zone, with strong agricultural traditions in tea, coffee, and horticulture. Kiambu's proximity to Nairobi makes it particularly urbanized, with a mixed economy spanning agriculture, industry, and services, while Murang'a and Kirinyaga retain a more predominantly rural character.

Selecting two contrasting three-county clusters one predominantly agricultural and rural (Rift Valley), the other agriculturally rooted but increasingly urbanizing (Central) — allows the study to examine whether NGCDF allocation patterns and equity outcomes differ systematically between regions with different economic and urbanization profiles.

1.2 Research Problem

The National Government Constituencies Development Fund represents a significant transfer of resources from the national government to the constituency level, with cumulative allocations exceeding hundreds of billions of shillings since its inception. Despite this substantial investment, limited systematic analysis exists regarding allocation patterns, intra-county equity, and inflation-adjusted growth, particularly when comparing counties from different regions of Kenya with distinct economic and demographic profiles.

Several challenges complicate the analysis of NGCDF allocations in the six study counties:

Data Fragmentation: NGCDF allocation data is often scattered across multiple sources, with inconsistent formats, variable quality, and limited accessibility, hampering comprehensive analysis of allocation patterns over time and across constituencies.

Nominal vs. Real Confusion: Public commentary on NGCDF allocations frequently cites nominal growth figures without adjusting for inflation, creating a misleading impression of funding growth when, in real terms, purchasing power may have grown far more modestly, or even declined.

Limited Cross-Regional Comparison: While individual county-level or national-level studies of NGCDF exist, few systematically compare allocation and equity patterns between two distinct multi-county regional clusters using a consistent analytical framework.

Equity Concerns: Questions persist regarding whether NGCDF allocations adequately reflect the diverse needs of different constituencies, especially where constituency counts, voter populations, and urbanization levels vary substantially within and across counties.

Accountability and Transparency: Despite legal requirements for transparency and public participation, accessing comprehensive, analysis-ready allocation data remains challenging for researchers, civil society organizations, and citizens, limiting opportunities for independent oversight.

This study addresses these challenges by developing a systematic, replicable analytical approach to NGCDF data analysis, focusing on six counties across two regions over a thirteen-year period (2014–2026), with forward-looking projections to 2030.

1.3 Research Objectives

The study was guided by the following objectives:

Main Objective: To analyze NGCDF allocation patterns, equity, and growth trajectories across Trans Nzoia, Nandi, Uasin Gishu (Group A) and Murang'a, Kiambu, Kirinyaga (Group B) counties for the period 2014–2030.

Specific Objectives:

1. To document allocation totals and year-on-year trends for NGCDF in the six study counties from 2014 to 2026, with linear projections through 2030.
2. To analyze constituency-level allocation distribution and classify constituencies into relative performance tiers within each county.
3. To assess intra-county equity (via Gini coefficients and coefficients of variation) and distinguish nominal from real (CPI-adjusted) allocation growth.
4. To provide a comparative, data-driven assessment of allocation patterns, per-voter funding, and national ranking between the Rift Valley (Group A) and Central (Group B) county clusters.

1.4 Research Questions

The study sought to answer the following questions:

1. What are the total NGCDF allocations to the six study counties for each year between 2014 and 2026, and what do linear projections suggest for 2027–2030?
2. How are allocations distributed across constituencies within each county, and which constituencies fall into high, mid, or low relative performance tiers?
3. How equitably are allocations distributed within each county, and how has this equity evolved over time?
4. To what extent does inflation erode the apparent gains suggested by nominal allocation growth?
5. How do the Rift Valley (Group A) and Central (Group B) county clusters compare in terms of total allocation, per-voter allocation, and national ranking?

1.5 Value of the Study

This study contributes value across multiple dimensions:

To Policymakers: The findings provide evidence-based insights into NGCDF allocation patterns and equity across agriculturally rooted and urbanizing county clusters, supporting informed decision-making regarding allocation formulas and monitoring frameworks.

To Development Practitioners: The study offers a replicable analytical framework combining trend analysis, equity metrics, per-voter normalization, and forecasting for monitoring constituency-level resource allocation across any set of Kenyan counties.

To Researchers: The research contributes to the academic literature on decentralized development and public finance management in Kenya, providing a methodological foundation for further research extending to additional counties or time periods.

To Citizens and Communities: By enhancing understanding of NGCDF allocation patterns and equity, the study supports citizen engagement in development planning and oversight, and can inform advocacy for more transparent and equitable resource allocation.

1.6 Scope and Limitations

The study focused on NGCDF allocations to Trans Nzoia, Nandi, Uasin Gishu, Murang'a, Kiambu, and Kirinyaga counties over the period 2014 to 2026, with linear projections extended to 2030. This scope was determined by data availability and the research objectives.

Inclusions:

- Analysis of annual NGCDF allocations at county and constituency levels
- Intra-county equity analysis (Gini coefficient, coefficient of variation)
- CPI-adjusted (real) allocation analysis using KNBS national inflation data
- Per-registered-voter allocation analysis using IEBC data
- National ranking of the six study counties against all 47 Kenyan counties
- Linear (OLS) forecasting of allocations through 2030

Exclusions:

- Analysis of project-level allocations and implementation details
- Examination of NGCDF-funded projects and their development outcomes
- Comparison with other development funds or government programs
- Primary data collection from NGCDF stakeholders or beneficiary communities

Limitations:

- The CPI index used is a national Kenya-wide measure and does not capture county-level price differences, which are particularly relevant for Kiambu given its proximity to Nairobi.
- Voter registration figures represent registered voters at a point in time and may not perfectly reflect the population of each constituency, particularly in rapidly urbanizing areas where migration and registration patterns may be out of sync.
- The analysis covers actual data up to 2026; figures for 2027–2030 are linear projections and should be interpreted as indicative rather than predictive.
- The study did not assess the impact or effectiveness of NGCDF-funded projects on development outcomes.

These limitations should be considered when interpreting the findings and recommendations presented in this report.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter presents a review of literature relevant to the analysis of NGCDF allocations in Kenya. The review covers theoretical foundations, a conceptual framework, empirical studies, and identified research gaps. The literature review draws on academic publications, government documents, and reports from development organizations to provide a comprehensive understanding of constituency development funds and resource allocation in decentralized contexts.

2.2 Theoretical Framework

The study is anchored on three theoretical perspectives that provide a foundation for understanding resource allocation in decentralized development contexts.

2.2.1 Public Finance Theory

Public finance theory, as articulated by Musgrave (Musgrave, 1959) and subsequently developed by Oates (Oates, 1972), provides fundamental insights into how governments should allocate resources to achieve economic efficiency, equity, and stability. The theory distinguishes between three primary functions of government:

Allocation Function: The provision of public goods and services that markets cannot efficiently provide. NGCDF contributes to this function by financing community-level public goods including schools, health facilities, water systems, and roads.

Distribution Function: The adjustment of income and wealth distribution to achieve equity objectives. NGCDF's allocation formula, with its poverty index component, reflects an effort to address distributional concerns by directing additional resources to disadvantaged constituencies.

Stabilization Function: The maintenance of economic stability through fiscal and monetary policy. While less directly relevant to constituency-level funds, the overall resource allocation framework contributes to regional economic stability.

Public finance theory suggests that decentralized resource allocation can improve efficiency by allowing local preferences and conditions to shape public service provision, while also recognizing the need for central oversight to ensure equity and maintain fiscal discipline (Oates, 1972).

2.2.2 Fiscal Decentralization Theory

Fiscal decentralization theory addresses the optimal assignment of revenue and expenditure responsibilities across levels of government. The theory, developed by scholars including Tiebout (Tiebout, 1956) and Oates (Oates, 1972), argues that decentralized service delivery can improve efficiency by allowing service levels to reflect local preferences, encouraging innovation in service delivery, enhancing accountability through closer proximity to citizens, and promoting competition among jurisdictions.

The NGCDF represents a form of fiscal decentralization where resources are transferred from the national government to constituency-level committees for local development priorities, aligning with the theoretical advantages of decentralization while maintaining central control over resource mobilization and overall allocation frameworks (Smoke, 2001). However, fiscal decentralization theory also identifies potential challenges relevant to NGCDF implementation, including capacity constraints at local levels, risk of capture by local elites, coordination challenges across jurisdictions, and potential for fiscal indiscipline.

2.2.3 Equity Theory in Resource Allocation

Equity theory addresses the fairness of resource distribution across populations and regions. In the context of public finance, equity can be understood through several lenses:

Horizontal Equity: Equal treatment of individuals or communities in similar circumstances. Applied to NGCDF, horizontal equity suggests that constituencies with similar characteristics should receive similar allocations per capita, or per registered voter.

Vertical Equity: Differential treatment to address different circumstances. The NGCDF allocation formula's poverty index component reflects vertical equity considerations by directing additional resources to disadvantaged constituencies.

Interregional Equity: Fair distribution of resources across geographic areas. This dimension is particularly relevant to a study comparing a predominantly rural, agricultural region (Rift Valley) against a more urbanizing region (Central).

Scholars including Rawls (Rawls, 1971) and Sen (Sen, 1992) have developed frameworks for understanding equity that go beyond simple equality of resources to consider capabilities, opportunities, and outcomes. These perspectives suggest that equitable resource allocation should consider not only the amount of resources provided, but also the capacity of communities to convert those resources into improved development outcomes — a consideration directly relevant to this study's use of Gini coefficients and coefficients of variation as monitoring tools rather than definitive verdicts of fairness.

2.3 Conceptual Framework

Based on the theoretical foundations discussed above, this study adopts a conceptual framework that links NGCDF allocations to administrative structure, time, and equity considerations.

Independent Variables:

- County administrative structure (number of constituencies)
- Registered voter population (IEBC data)
- Time (financial year of allocation)
- Regional grouping (Rift Valley vs. Central)

Dependent Variables:

- Total NGCDF allocation (county and constituency level)
- Per-voter allocation
- Year-on-year and compound growth rates (nominal and real)
- Intra-county Gini coefficient and coefficient of variation

Intervening Variables:

- NGCDF allocation formula (equal share, population, poverty index, emergency reserve)
- National budget processes and fiscal conditions
- Consumer Price Index / inflation

The framework posits that observed allocation patterns result from the interaction of these variables, with the allocation formula mediating between constituency characteristics and actual allocations, and inflation mediating between nominal allocation figures and their real purchasing power. By analyzing allocation patterns alongside equity metrics and inflation-adjusted figures, the study seeks to understand the extent to which allocations reflect both formula-driven realities and genuine gains in development purchasing power.

2.4 Empirical Literature Review

2.4.1 Constituency Development Funds in Kenya

Several studies have examined the operation and impact of constituency development funds in Kenya. Bagaka (Bagaka, 2009) analyzed the fund's role in promoting development at the local level, finding that while the fund had financed numerous projects, its impact was constrained by implementation challenges including limited community participation and weak project monitoring.

Kimenyi (Kimenyi, 2005) examined the evolution of the fund from its establishment through subsequent amendments, highlighting the tension between centralized oversight and local autonomy. Mwangi and Otieno (Mwangi & Otieno, 2022) conducted a comprehensive analysis of NGCDF allocations across Kenyan constituencies, identifying significant variations in per-capita and per-voter allocations, and raising questions about the equity of the allocation formula. Odhiambo (Odhiambo, 2021) examined the governance structures of the fund, focusing on the role of constituency committees in project identification and oversight, finding that Members of Parliament and local elites often dominated decision-making processes in practice.

2.4.2 NGCDF Allocation Patterns

Research on NGCDF allocation patterns has revealed several important findings. The Kenya National Bureau of Statistics (Kenya National Bureau of Statistics, 2020) documented significant variations in allocations across counties and constituencies, with urban constituencies generally receiving higher total allocations but lower per-capita allocations compared to rural constituencies — a pattern directly relevant to this study's comparison of urbanizing Kiambu against its more rural Central-region peers.

Cheeseman, Kanyinga, and Lynch (Cheeseman et al., 2020) examined the political economy of NGCDF allocations, finding evidence that allocations may be influenced by political considerations including the proximity of elections and the political alignment of constituency representatives. This finding raises important questions about the extent to which allocations follow the prescribed formula versus responding to political dynamics — a factor outside the scope of this study's quantitative analysis but relevant to interpreting its findings.

2.4.3 Per-Voter and Equity Analysis of Public Funds

Per-capita and per-voter analysis has emerged as an important tool for assessing the equity of public resource allocation. The Commission on Revenue Allocation (Commission on Revenue Allocation, 2022) has developed sophisticated models for assessing the equitable distribution of resources across counties, incorporating factors including population, poverty, land area, and development indicators, though these models have not been systematically applied to constituency-level analysis. Studies by the World Bank (World Bank, 2020) and USAID (United States Agency for International Development, 2021) have similarly used per-capita and related metrics to compare resource allocation across regions and identify potential inequities.

The application of per-voter analysis to NGCDF allocations, as used in this study, addresses a practical gap identified in prior research: the lack of consistently available, constituency-level population data. Registered voter counts, while an imperfect proxy for total population, provide a consistently available, temporally comparable base for normalizing allocation figures across constituencies of different sizes.

2.5 Research Gaps

The literature review reveals several gaps that this study seeks to address:

Inflation-Adjustment Gap: Much existing research on NGCDF reports nominal allocation figures without systematically adjusting for inflation. This study addresses this gap through a dedicated CPI-adjustment methodology that distinguishes nominal from real growth across the full study period.

Cross-Regional Comparison Gap: While individual county studies exist, limited research has systematically compared allocation and equity patterns between two distinct, multi-county regional clusters using a consistent analytical framework. This study's Rift Valley versus Central comparison addresses this gap.

Intra-County Equity Monitoring Gap: Despite the allocation formula's equity objectives, limited research has systematically tracked intra-county distributional equity over time using standardized metrics such as the Gini coefficient and coefficient of variation. This study's longitudinal equity analysis contributes to filling this gap.

Forecasting Gap: Existing NGCDF literature is largely retrospective. This study's use of OLS-based linear forecasting to project allocations through 2030 provides a forward-looking planning tool largely absent from prior academic work.

Methodological Gap: The literature reveals limited application of modern statistical and visualization techniques, including the R programming language, to NGCDF data. This study contributes to methodological development in this area by combining trend analysis, equity metrics, inferential statistical tests, and forecasting within a single reproducible framework.

By addressing these gaps, the study contributes to both academic understanding and practical application of NGCDF analysis.

CHAPTER THREE: RESEARCH METHODOLOGY

3.1 Introduction

This chapter presents the methodology employed in conducting the NGCDF allocation analysis. The chapter covers the research design, data sources and collection methods, data processing and cleaning procedures, the CPI-adjustment and forecasting methodologies, and the equity metrics and statistical tests applied. The methodology was designed to ensure rigorous, replicable analysis while acknowledging the limitations inherent in secondary data analysis.

3.2 Research Design

The study employed a quantitative research design with descriptive, comparative, and inferential components. The descriptive component documented allocation totals, trends, and constituency-level distribution patterns across the six study counties and their constituent constituencies. The comparative component contrasted the Rift Valley (Group A) and Central (Group B) regional clusters. The inferential component applied statistical tests to assess whether observed differences across counties and years were statistically significant.

The research design was guided by the following considerations:

Temporal Coverage: The study covered the thirteen-year period from 2014 to 2026 for actual data, with linear projections extending the analysis to 2030.

Geographic Scope: The study focused on six counties organized into two regional clusters of three counties each — Trans Nzoia, Nandi, and Uasin Gishu (Group A); and Murang'a, Kiambu, and Kirinyaga (Group B) — selected for their contrasting regional and economic profiles.

Unit of Analysis: The primary units of analysis were counties and constituencies, enabling examination of allocation patterns at both levels.

Data Sources: The study relied on secondary data from official sources (NGCDF Board, IEBC, KNBS), ensuring reliability and enabling replication.

3.3 Data Sources and Collection

The analysis is based on official NGCDF constituency allocation data for the period 2014–2026, sourced from the **NGCDF Board** (National Government Constituencies Development Fund Board, 2025). The data set covers all parliamentary constituencies within the six study counties, with annual allocation figures recorded for each financial year. Voter registration figures used for per-voter computations are drawn from the **Independent Electoral and Boundaries Commission (IEBC)** registered voter rolls (Independent Electoral and Boundaries Commission, 2022).

3.3.1 Data Cleaning and Preparation

The raw data set underwent the following cleaning steps before analysis:

1. **Column name standardization:** Year columns originally prefixed with `X` (e.g., `X2014`) were stripped of the prefix and converted to plain integer year labels.
2. **County name normalization:** The county name "Murang'a" appears in multiple forms in the raw data (with and without the apostrophe, and with varying capitalization). All variants were harmonized to the standard form `Murang'a` using a case-insensitive pattern match.
3. **Numeric conversion:** All allocation and voter columns were coerced to numeric type, with warnings suppressed for non-numeric values that were treated as missing (`NA`).
4. **Missing data handling:** Rows with missing county or constituency identifiers were excluded. Year-constituency combinations with missing allocation values were excluded from aggregations for those specific years but did not cause the constituency to be dropped from all analyses.

3.3.2 Limitations of the Data

- The CPI index used is a national Kenya-wide measure and does not capture county-level price differences, which are particularly relevant for Kiambu given its proximity to Nairobi.
- Voter registration figures represent registered voters at a point in time and may not perfectly match the population of each constituency, particularly in rapidly urbanizing areas where migration and registration patterns may be out of sync.
- The analysis covers the period up to 2026; figures for 2026 may in some cases be preliminary or estimated depending on the data extraction date.

3.4 CPI Adjustment Methodology

All CPI adjustments use Kenya's **Consumer Price Index** (Kenya National Bureau of Statistics, 2025) with 2014 as the base year (index = 100). The deflator series (table below) allows nominal allocations to be converted to constant 2014 Kenya Shillings, enabling meaningful inter-temporal comparisons of purchasing power.

Table 1: Kenya CPI deflator series used in this analysis (2014 = 100)

Year	CPI Index	Annual Inflation (%)
2014	100	NA
2015	106	6.0
2016	111	4.7
2017	118	6.3
2018	126	6.8
2019	131	4.0
2020	138	5.3
2021	148	7.2
2022	162	9.5
2023	177	9.3
2024	189	6.8
2025	198	4.8
2026	208	5.1

3.4.1 Why Inflation Adjustment Matters

Without CPI adjustment, comparing a 2025 allocation to a 2014 allocation in raw KES terms is like comparing apples to oranges — the 2025 shilling is worth significantly less than the 2014 shilling. Kenya experienced cumulative inflation of approximately 77% between 2014 and 2025, meaning that KES 100 in 2025 buys roughly the same as KES 56 did in 2014.

For NGCDF analysis, this matters in two ways. First, it affects the real value of allocations. A constituency whose allocation grew 50% nominally over the period actually received less in real terms than it did at the start, because inflation outpaced its funding growth. Second, it affects the design and cost of development projects: a school classroom or borehole that cost KES 2 million to build in 2014 may cost KES 3.5 million or more today, meaning a constituency needs its allocation to grow significantly just to maintain the same construction output.

The CPI series used in this report is reproduced below, along with the implied annual inflation rate, for full transparency.

```
CPI_DF %>%
  mutate(pct_change = round((cpi / lag(cpi) - 1) * 100, 1)) %>%
  rename(Year = year, `CPI Index` = cpi, `Annual Inflation (%)` = pct_change) %>%
  kbl(align = c("c", "r", "r")) %>%
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover"))
```

3.5 Forecasting Approach

County-level forecasts for 2027–2030 are generated using **ordinary least squares (OLS) linear regression**, with the 2020–2026 period as the estimation window to capture post-COVID-19 fiscal trajectory. The model form is:

$$\text{Total Allocation}_t = \beta_0 + \beta_1 \cdot t + \varepsilon_t$$

Forecast confidence intervals are not reported given the exploratory nature of the projections and the short estimation window. Projections should be interpreted as **indicative rather than predictive**.

3.6 Equity Metrics

Two complementary measures of intra-county distributional equity are reported:

- **Gini Coefficient:** Ranges from 0 (perfect equality) to 1 (maximum inequality). Computed annually for each county using constituency-level allocation data via the `ineq` package.
- **Coefficient of Variation (CV):** Standard deviation expressed as a percentage of the mean. Provides a scale-invariant measure of relative dispersion.

3.7 Statistical Tests

- **One-way ANOVA:** Tests whether mean 2025 constituency allocations differ significantly across the counties within each group.
- **Kruskal-Wallis test:** Non-parametric equivalent of ANOVA; used to corroborate findings where normality may be violated.
- **Paired t-test (2024 vs 2025):** Tests whether the mean constituency allocation changed significantly between the two most recent years.
- **Pearson correlation:** Measures the linear association between registered voter count and 2025 allocation at the constituency level.

CHAPTER FOUR: DATA ANALYSIS, FINDINGS AND DISCUSSION

4.1 Introduction

This chapter presents the findings from the analysis of NGCDF allocations across the six study counties for the period 2014–2026, with linear projections through 2030. The chapter is organized according to the research objectives: Section 4.2 presents the Group A (Rift Valley) analysis, covering allocation trends and forecasts, cumulative allocation, constituency performance tiers, intra-county equity, CPI-adjusted allocations, per-voter allocation, national ranking, comparison with neighbouring counties, and statistical tests. Section 4.3 mirrors this structure for Group B (Central). Section 4.4 then brings all six counties together for a direct cross-group comparison. Section 4.5 closes the chapter with a discussion synthesizing the findings across both groups.

4.2 Group A: Rift Valley — Trans Nzoia, Nandi & Uasin Gishu

The three Rift Valley counties in this group, Trans Nzoia, Nandi, and Uasin Gishu, collectively cover a significant portion of Kenya's western agricultural heartland. Trans Nzoia is famed for large-scale maize production and is often called Kenya's "bread basket." Nandi is characterized by its tea-growing highlands and is home to some of Kenya's most celebrated long-distance runners. Uasin Gishu, centered on Eldoret, the fifth largest city in Kenya, has a more diversified economy with significant manufacturing, educational institutions (including Moi University), and the country's second busiest international airport. Despite geographic proximity, the three counties differ in number of constituencies (Trans Nzoia has 5, Nandi has 6, and Uasin Gishu has 6), which means that headline county-level totals must be read alongside per-constituency and per-voter figures to avoid misleading comparisons.

4.2.1 Allocation Trends and Forecasts

Understanding how NGCDF allocations have evolved over time is fundamental to assessing whether development funding has kept pace with population growth, inflation, and the expanding developmental needs of constituencies. The chart below plots the annual total allocation for each county from 2014 to 2026, followed by a linear projection to 2030.

The trend line is fitted using Ordinary Least Squares (OLS) regression on the 2020–2026 sub-period, which was chosen deliberately to exclude the earlier slower-growth years and reflect the more recent trajectory of allocations. A vertical dashed line marks the boundary between observed actual and projected values, making it easy to distinguish the two.

Key things to look for in this chart: Is one county's trajectory steeper (faster-growing) than others? Are there any noticeable breaks or plateaus in the trend, for instance, around 2020–2021, which corresponds to the COVID-19 pandemic period when national revenues were under pressure? How do the three counties compare in absolute terms, and what does that imply about the size of their respective constituency bases?

```r
trend_A <- cy_A %>% select(county, year, total) %>% mutate(type = "Actual") %>% bind_rows(fc_
ggplot(trend_A, aes(x = year, y = total / 1e9,
                    colour = county,
                    linetype = type,
                    group = interaction(county, type))) +
  geom_vline(xintercept = 2026.5, colour = "#94a3b8", linetype = "dashed", linewidth = 0.6) +
  geom_line(linewidth = 1.5) +
  geom_point(data = ~ filter(.x, type == "Actual"),
             size = 2.5, shape = 21, fill = "white", stroke = 2) +
  geom_point(data = ~ filter(.x, type == "Forecast"),
             size = 3, shape = 17) +
  annotate("text", x = 2027.3, y = max(trend_A$total / 1e9) * 0.55,
           label = "Forecast →", colour = "#6366f1", size = 3.2, fontface = "italic") +
  scale_colour_manual(values = COL) +
  scale_linetype_manual(values = c("Actual" = "solid", "Forecast" = "dashed")) +
  scale_x_continuous(breaks = c(2014, 2017, 2020, 2023, 2026, 2027, 2030)) +
  scale_y_continuous(labels = label_number(suffix = "B", prefix = "KES ")) +
  labs(
    title    = "Group A: NGCDF County Totals Actuals + Forecast 2027-2030",
    subtitle = "Linear trend fitted on 2020-2026 trajectory",
    x = NULL, y = "Total (KES Billion)", colour = NULL, linetype = NULL) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1), legend.position = "top")
```

Table 2: Group A — Projected NGCDF allocations 2027–2030 (KES, rounded)

	Projected Allocation (KES)		
Year	Nandi	Trans Nzoia	Uasin Gishu
2027	1,168,474,596	973,711,377	1,168,469,345
2028	1,225,591,539	1,021,305,044	1,225,584,026
2029	1,282,708,482	1,068,898,711	1,282,698,706
2030	1,339,825,426	1,116,492,378	1,339,813,386

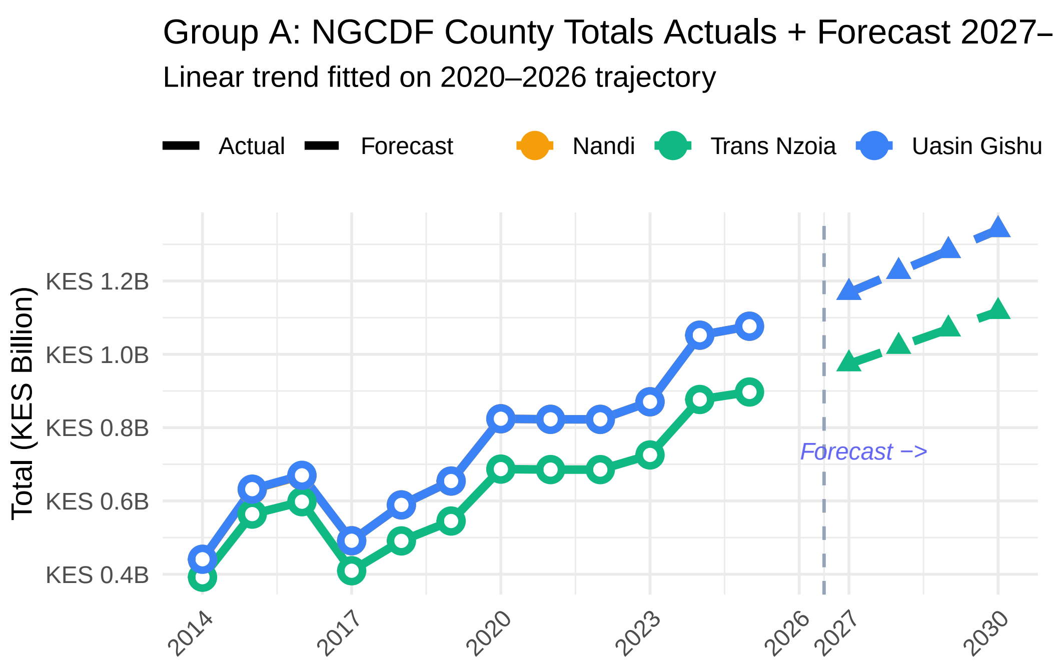

Figure 1: Group A: Annual NGCDF county totals (actuals 2014–2026) and linear forecast 2027–2030. Dashed lines indicate projected values.

```
fc_A %>%
  mutate(total = round(total)) %>%
  pivot_wider(names_from = county, values_from = total, id_cols = year) %>%
  rename(Year = year) %>%
  mutate(across(-Year, ~ format(.x, big.mark = ","))) %>%
  kbl(align = c("l", rep("r", 3))) %>%
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover", "condensed")) 
  add_header_above(c(" " = 1, "Projected Allocation (KES)" = 3))
```

The forecast table above translates the graphical projections into precise figures. These numbers should be interpreted as the expected allocation **if current trends continue uninterrupted** they are not guarantees, and significant national budget shifts, changes in the NGCDF formula, or economic shocks could cause actual allocations to deviate materially from these projections. The figures are most useful as planning benchmarks: county governments and constituencies can use them to estimate future resource envelopes for multi-year development planning.

4.2.2 Cumulative Allocation

While annual figures show year-to-year fluctuations, **cumulative allocation** provides a more complete picture of the total development investment that a county has received (or is projected to receive) over the entire period. A county that received lower allocations in early years but has been growing faster may ultimately accumulate more resources than a county that started higher but grew more slowly.

The area chart below shows the running cumulative total for each county from 2014 through to the 2030 projection horizon. The shaded area under each county's curve (for the actual period) helps visually communicate the total volume of funds. The dashed lines extending beyond 2026 represent the projected cumulative growth based on the linear forecast.

```
trend_A %>% arrange(county, year) %>% group_by(county) %>%
  mutate(cumul = cumsum(total)) %>% ungroup() %>%
  ggplot(aes(x = year, y = cumul / 1e9,
             fill = county, colour = county,
             linetype = type,
             group = interaction(county, type))) +
  geom_area(data = ~ filter(.x, type == "Actual"), alpha = 0.15, position = "identity") +
  geom_line(linewidth = 1.4) +
  scale_colour_manual(values = COL) +
  scale_fill_manual(values = COL) +
  scale_linetype_manual(values = c("Actual" = "solid", "Forecast" = "dashed")) +
  scale_x_continuous(breaks = c(2014, 2018, 2022, 2026, 2030)) +
  scale_y_continuous(labels = label_number(suffix = "B", prefix = "KES ")) +
  labs(
    title    = "Group A: Cumulative NGCDF Allocation 2014-2030",
    subtitle = "Running total per county, including projected years",
    x = NULL, y = "Cumulative (KES Billion)", colour = NULL, linetype = NULL) + theme(legend
```

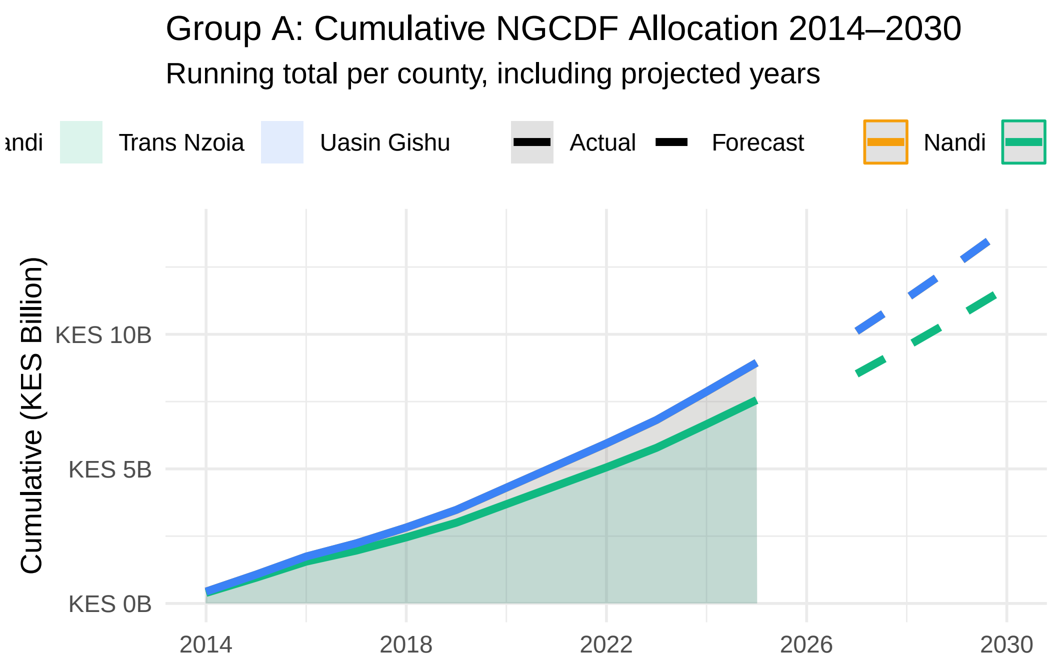

Figure 2: Group A: Cumulative NGCDF allocation 2014–2030 (including projections).

4.2.3 Constituency Performance Tiers (2025)

Not all constituencies within a county receive the same NGCDF allocation. While the fund's formula is designed to be broadly equitable, differences in constituency size, number of registered voters, and historical patterns can result in some constituencies receiving meaningfully more than others within the same county. To make these differences visible, each constituency is classified into one of three performance tiers based on where its 2025 allocation falls within the county's own distribution:

- **High tier** (green): Allocations in the top third of the county's distribution, these constituencies received relatively more funding compared to their county peers.
- **Mid tier** (amber): Allocations in the middle third broadly average within the county.
- **Low tier** (red): Allocations in the bottom third, these constituencies received comparatively less than the county average.

This classification is entirely **relative and intra-county** — a "Low" tier constituency in one county may still receive more in absolute terms than a "High" tier constituency in another county. The purpose is not to flag underperformance in any absolute sense, but to highlight where within-county disparities exist and which constituencies may benefit most from targeted attention.

The `vs County Mean (%)` column in the accompanying table shows how far above or below the county average each constituency sits, providing a precise measure of the distributional gap. The `Per Voter (KES)` column further contextualizes each constituency's allocation by expressing it per registered voter, which is arguably the most equitable metric of all since it normalizes for differences in constituency population size.

Constituencies are classified into **High**, **Mid**, and **Low** tiers based on intra-county tertiles of the 2025 allocation.

```
tier_A %>%
  ggplot(aes(x = `2025` / 1e6, y = reorder(constituency, `2025`), fill = tier)) +
  geom_col(width = 0.75) +
  geom_text(aes(label = tier), hjust = -0.2, size = 2.8, colour = "#475569") +
  scale_fill_manual(values = c("High" = "#10b981", "Mid" = "#f59e0b", "Low" = "#ef4444")) +
  scale_x_continuous(
    labels = label_number(suffix = "M", prefix = "KES "),
    expand = expansion(mult = c(0, 0.25))) + facet_wrap(~ county, scales = "free_y", ncol =
  labs(
    title    = "Group A: Constituency Performance Tiers 2025",
    subtitle = "High = top third · Mid = middle third · Low = bottom third",
    x = "Allocation (KES Millions)", y = NULL, fill = "Tier") + theme(legend.position = "top
```

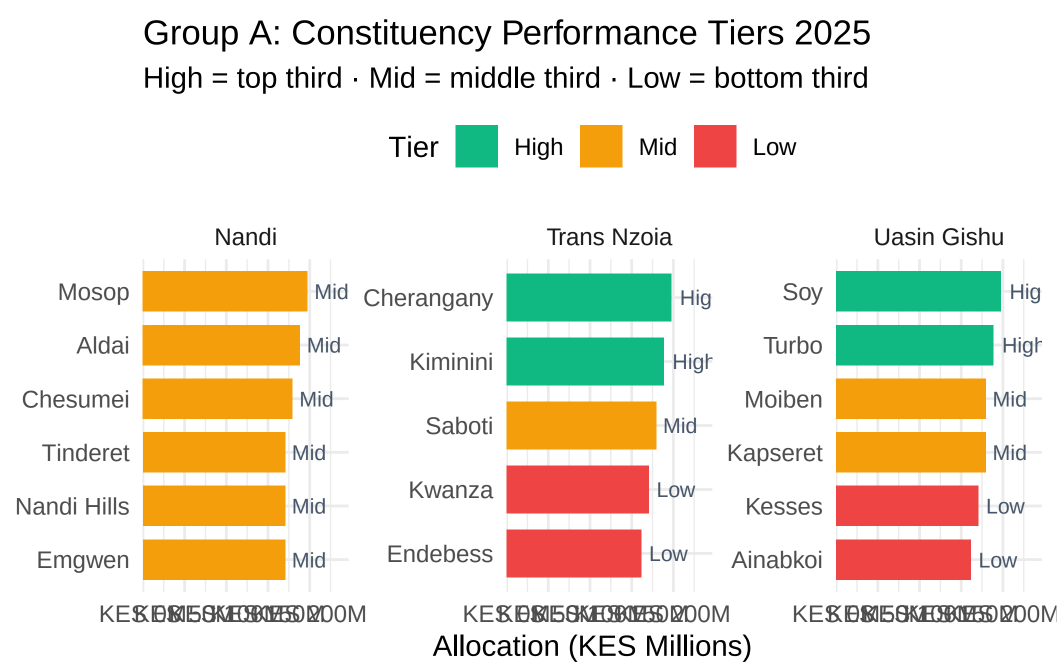

Figure 3: Group A: Constituency-level 2025 allocations colour-coded by performance tier (High / Mid / Low).

```r
tier_A %>%
  select(County = county, Constituency = constituency,
         Voters = voters, `Allocation (KES)` = `2025`,
         Tier = tier, `vs County Mean (%)` = vs_mean,
         `Per Voter (KES)` = per_voter) %>%
  arrange(County, desc(Tier)) %>%
  mutate(
    `Allocation (KES)`   = format(round(`Allocation (KES)`), big.mark = ","),
    `vs County Mean (%)` = round(`vs County Mean (%)`, 1),
    `Per Voter (KES)`    = format(round(`Per Voter (KES)`), big.mark = ","),
    Voters               = format(Voters, big.mark = ",")
  ) %>%
  kbl(align = c("l", "l", "r", "r", "c", "r", "r")) %>%
  kable_styling(full_width = TRUE, bootstrap_options = c("striped", "hover", "condensed"),
                font_size = 11) %>%
  column_spec(5, bold = TRUE,
              color = ifelse(tier_A %>% arrange(county, desc(tier)) %>% pull(tier) == "High"
                      ifelse(tier_A %>% arrange(county, desc(tier)) %>% pull(tier) == "Mid",
                             "#ef4444"))) %>%
  collapse_rows(columns = 1, valign = "top")
```

4.2.4 Equity and Intra-County Distribution (Gini)

While the tier analysis provides a snapshot of 2025 distribution, tracking equity over time reveals whether a county is becoming more or less equitable in how it distributes NGCDF funds across its constituencies. This section uses two complementary statistical measures:

Gini Coefficient

The **Gini coefficient** is a well-established measure of inequality, most commonly used in economics to measure income distribution but equally applicable to any distributional question. It ranges from 0 (perfect equality, every constituency receives exactly the same allocation) to 1 (maximum inequality, one constituency receives everything and the rest receive nothing). In practice, NGCDF Gini coefficients will typically fall between 0.05 and 0.40.

A rising Gini over time indicates that allocations within the county are becoming more concentrated among a smaller number of constituencies, potentially a concern for equitable development. A declining Gini suggests convergence, with the gap between the best- and worst-funded constituencies narrowing.

It is important to note that some degree of variation is expected and even justifiable, constituencies with larger voter populations or greater infrastructure deficits may legitimately

Table 3: Group A — Constituency tier classification and per-voter allocation (2025)

County	Constituency	Voters	Allocation (KES)	Tier	vs County Mean (%)	Per Voter (KES)
Nandi	Tinderet	51,446	170,469,857	Mid	-5	3,314
	Aldai	78,005	188,414,052	Mid	5	2,415
Trans Nzoia	Nandi Hills	57,910	170,469,857	Mid	-5	2,944
	Chesumei	74,201	179,441,954	Mid	0	2,418
	Emgwen	66,940	170,469,857	Mid	-5	2,547
	Mosop	77,786	197,386,150	Mid	10	2,538
	Kwanza	75,839	170,469,857	Low	-5	2,248
	Endebess	50,688	161,497,759	Low	-10	3,186
	Saboti	87,384	179,441,954	Mid	0	2,053
	Kiminini	93,240	188,414,052	High	5	2,021
	Cherangany	91,830	197,386,150	High	10	2,149
Uasin Gishu	Ainabkoi	62,606	161,497,759	Low	-10	2,580
	Kesses	77,942	170,469,857	Low	-5	2,187
	Moiben	77,877	179,441,954	Mid	0	2,304
	Kapseret	74,809	179,441,954	Mid	0	2,399
	Soy	92,702	197,386,150	High	10	2,129
	Turbo	120,202	188,414,052	High	5	1,567

receive more. The Gini should therefore be interpreted as a **monitoring tool** rather than a verdict on the fund's fairness.

Coefficient of Variation (CV)

The **Coefficient of Variation (CV)** expresses the standard deviation of constituency allocations as a percentage of the mean. Unlike the Gini, which focuses on the shape of the entire distribution, the CV is particularly sensitive to extreme values at the top and bottom. A high CV indicates that allocations are spread widely around the mean, while a low CV suggests they cluster tightly around it.

Together, the Gini and CV provide a robust picture of distributional dynamics over time. If both are rising, it strongly suggests increasing inequality within the county. If they diverge, it may indicate a shift in the structure of inequality (e.g., more extreme outliers without a change in the middle of the distribution).

```
pE1_A <- gini_A %>%
  ggplot(aes(x = year, y = gini, colour = county, group = county)) +
  geom_line(linewidth = 1.6) +
  geom_point(size = 3, shape = 21, fill = "white", stroke = 2) +
  scale_colour_manual(values = COL) + scale_x_continuous(breaks = 2014:2026) + scale_y_conti
    title    = "Group A: Intra-County Gini - 2014 to 2026",
    subtitle = "0 = perfectly equal · Higher = more unequal",
    x = NULL, y = "Gini Coefficient", colour = NULL) + theme(axis.text.x = element_text(angl
pE2_A <- gini_A %>%
  ggplot(aes(x = year, y = cv, colour = county, group = county)) +
  geom_line(linewidth = 1.6) +
  geom_point(size = 3, shape = 21, fill = "white", stroke = 2) +
  scale_colour_manual(values = COL) +
  scale_x_continuous(breaks = 2014:2026) +
  scale_y_continuous(labels = label_percent(scale = 1)) +
  labs(
    title = "Group A: Coefficient of Variation (CV)",
    x = NULL, y = "CV (%)", colour = NULL) + theme(axis.text.x = element_text(angle = 45, hju
pE2_A / pE2_A
```

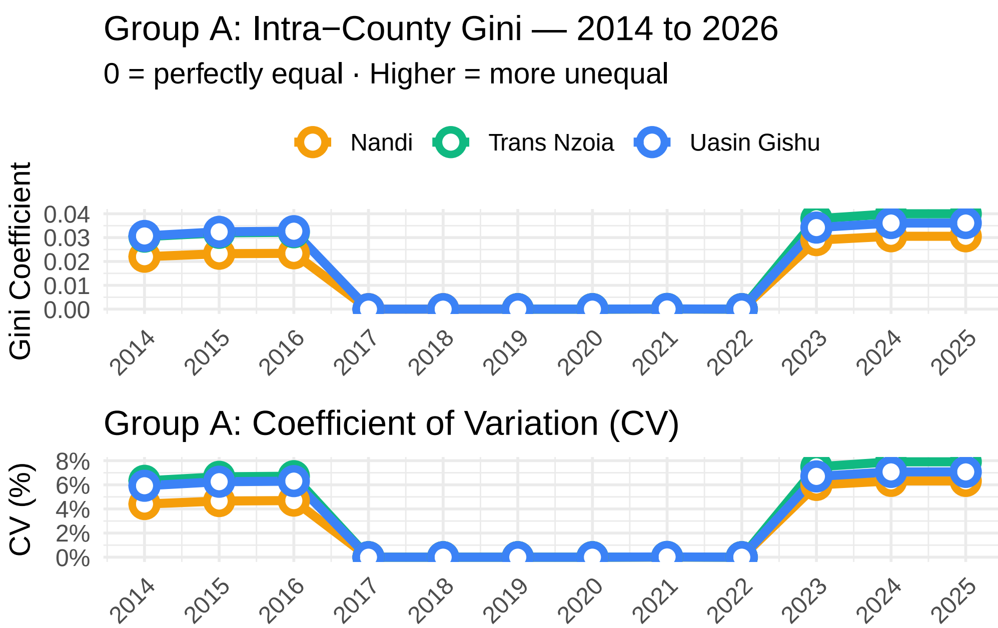

Figure 4: Group A — Gini coefficient (top) and Coefficient of Variation (bottom) measuring intra-county allocation inequality, 2014–2026.

```r
gini_A %>%
  filter(year == 2025) %>%
  mutate(
    gini          = round(gini, 4),
    cv            = round(cv, 2),
    max_min_ratio = round(max_min_ratio, 3)) %>%
  rename(County = county, Year = year,
         `Gini Coefficient` = gini,
         `CV (%)` = cv,
         `Max/Min Ratio` = max_min_ratio) %>%
  kbl(align = c("l", "c", "r", "r", "r")) %>%
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover"))
```

4.2.5 CPI-Adjusted (Real) Allocations

One of the most important, and often overlooked, aspects of public expenditure analysis is distinguishing between **nominal** and **real** changes in funding. A county that receives 20% more in 2025 than in 2014 in nominal terms has not necessarily received 20% more in terms

Table 4: Group A: Gini, CV, and max/min ratio for 2025

County	Year	Gini Coefficient	CV (%)	Max/Min Ratio
Nandi	2025	0.0306	6.32	1.158
Trans Nzoia	2025	0.0400	7.91	1.222
Uasin Gishu	2025	0.0361	7.07	1.222

of what those funds can actually buy. If inflation over the same period was, say, 77% (as has been the case in Kenya from 2014 to 2025), then a 20% nominal increase actually represents a significant decline in real purchasing power.

This distinction matters enormously for development planning. If a constituency's allocation has grown by KES 5 million since 2014 but construction costs, materials, and labor have all risen by over 70%, the constituency may find that its 2025 allocation builds far fewer classrooms or boreholes than the same nominal amount would have in 2014.

How the Adjustment Works

This report uses Kenya's **Consumer Price Index (CPI)** to deflate nominal allocations into constant 2014 prices. The formula applied is:

$$\text{Real Allocation}_t = \frac{\text{Nominal Allocation}_t}{\text{CPI}_t/100}$$

Where CPI is expressed with 2014 = 100. This produces a figure in 2014 purchasing power, allowing consistent comparisons across time.

Index Chart Interpretation

The chart below presents allocations as an **index (2014 = 100)** for both nominal and real terms. When a county's real index line is significantly below its nominal line, inflation has eroded a large share of the apparent funding gains. The dotted horizontal line at 100 represents the 2014 baseline, any county whose real index falls below this line has, in real terms, received less than it did at the start of the series.

```
cpi_A <- cy_A %>%
  select(county, year, total, real_total) %>%
  mutate(
    nominal_idx = total      / total[year == 2014]      * 100,
    real_idx    = real_total / real_total[year == 2014] * 100,
    .by = county)
cpi_A %>%
```

```
  pivot_longer(c(nominal_idx, real_idx), names_to = "type", values_to = "idx") %>%
  mutate(type = recode(type,
    "nominal_idx" = "Nominal (current KES)",
    "real_idx"    = "Real (2014 KES)")) %>%
  ggplot(aes(x = year, y = idx, colour = county, linetype = type)) +
  geom_hline(yintercept = 100, colour = "#94a3b8", linetype = "dotted") +
  geom_line(linewidth = 1.4) +
  geom_point(size = 2) +
  scale_colour_manual(values = COL) +
  scale_linetype_manual(values = c("Nominal (current KES)" = "solid", "Real (2014 KES)" = "da
  scale_x_continuous(breaks = c(2014, 2017, 2020, 2023, 2026)) +
  facet_wrap(~ county, ncol = 3) +
  labs(
    title    = "Group A: Nominal vs CPI-Adjusted Allocation Index",
    subtitle = "Base 2014 = 100",
    x = NULL, y = "Index (2014 = 100)", colour = NULL, linetype = NULL) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1), legend.position = "top")
```

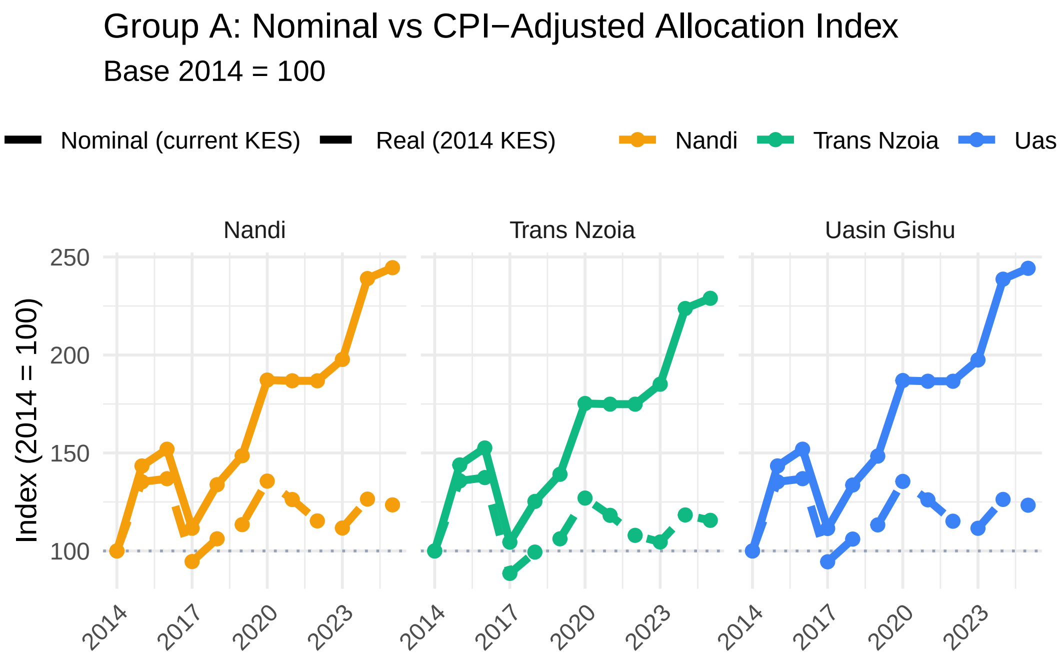

Figure 5: Group A: Nominal vs CPI-adjusted (real) allocation index, 2014 = 100. Dashed lines represent purchasing-power-adjusted values.

Table 5: Group A: Nominal vs real (2014 KES) allocation comparison for 2025

County	Nominal (KES)	Real 2014 KES	Inflation Gap (%)
Nandi	1,076,651,727	543,763,498	98
Trans Nzoia	897,209,772	453,136,249	98
Uasin Gishu	1,076,651,727	543,763,498	98

Note:

Kenya CPI growth 2014–2025: 98%

```
cpi_growth <- (CPI_DF$cpi[CPI_DF$year == 2025] / CPI_DF$cpi[CPI_DF$year == 2014] - 1) * 100
cy_A %>%
  filter(year == 2025) %>%
  select(county, nominal = total, real = real_total) %>%
  mutate(
    pct_difference = round((nominal / real - 1) * 100, 1),
    nominal_growth = round((nominal /
      (cy_A %>% filter(year == 2014) %>%
         filter(county %in% GROUP_A) %>%
         select(county, total_2014 = total) %>%
         right_join(tibble(county = GROUP_A), by = "county") %>%
         pull(total_2014)) - 1) * 100, 1),
    across(c(nominal, real), ~ format(round(.x), big.mark = ","))) %>%
  rename(County = county,
         `Nominal (KES)` = nominal,
         `Real 2014 KES` = real,
         `Inflation Gap (%)` = pct_difference) %>%
  select(County, `Nominal (KES)`, `Real 2014 KES`, `Inflation Gap (%)`) %>%
  kbl(align = c("l", "r", "r", "r")) %>%
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover")) %>%
  footnote(general = paste0("Kenya CPI growth 2014-2025: ", round(cpi_growth, 1), "%"))
```

4.2.6 Per-Voter Allocation and Voter Context

Comparing absolute allocation totals between constituencies can be misleading when their population sizes differ substantially. A constituency with 100,000 registered voters receiving KES 80 million may actually be less well funded than one with 40,000 voters receiving KES 50 million — because in the former case the allocation per voter is only KES 800, while in the latter it is KES 1,250.

Per-voter allocation the total constituency NGCDF allocation divided by the number of registered voters — corrects for this by providing a population-normalized measure of funding intensity. It is arguably the most equitable metric in this analysis, asking: how much development funding is each voter's constituency receiving on their behalf?

What the Charts Show

The **upper panel** tracks how per-voter allocation has evolved over time at the county level. A rising trend indicates that funding growth has outpaced voter registration growth, a positive sign for development equity. A flat or declining trend may mean that voter rolls are expanding faster than allocations, effectively diluting the per-voter resource envelope even if nominal totals are rising.

The **lower panel** maps individual constituencies by their 2025 allocation (y-axis) against the number of registered voters (x-axis), with each point labelled by constituency name. The fitted regression line shows the overall relationship. Constituencies falling significantly above the line are receiving more than expected for their voter count; those below receive less.

```r
pV1_A <- cy_A %>%
  ggplot(aes(x = year, y = per_voter, colour = county, group = county)) +
  geom_line(linewidth = 1.6) +
  geom_point(size = 3, shape = 21, fill = "white", stroke = 2) +
  scale_colour_manual(values = COL) +
  scale_x_continuous(breaks = 2014:2026) +
  scale_y_continuous(labels = label_number(big.mark = ",", prefix = "KES ")) +
  labs(
    title = "Group A: Allocation Per Registered Voter 2014 to 2026",
    x = NULL, y = "KES per Voter", colour = NULL) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1), legend.position = "top")
pV2_A <- df_A %>%
  filter(!is.na(`2025`), !is.na(voters), voters > 0) %>%
  ggplot(aes(x = voters / 1000, y = `2025` / 1e6, colour = county, label = constituency)) +
  geom_smooth(aes(group = 1), method = "lm", colour = "#64748b",
              se = TRUE, fill = "#cbd5e1", alpha = 0.3, linewidth = 1) +
  geom_point(size = 4, alpha = 0.85) +
  geom_text(size = 2.8, colour = "#475569", vjust = -1, check_overlap = TRUE) +
  scale_colour_manual(values = COL) +
  scale_x_continuous(labels = label_number(suffix = "K")) +
  scale_y_continuous(labels = label_number(suffix = "M", prefix = "KES ")) +
  labs(
    title = "Group A: Allocation vs Voter Population - 2025",
    x = "Registered Voters (Thousands)", y = "Allocation (KES Millions)", colour = NULL) + th
pV1_A / pV2_A
```

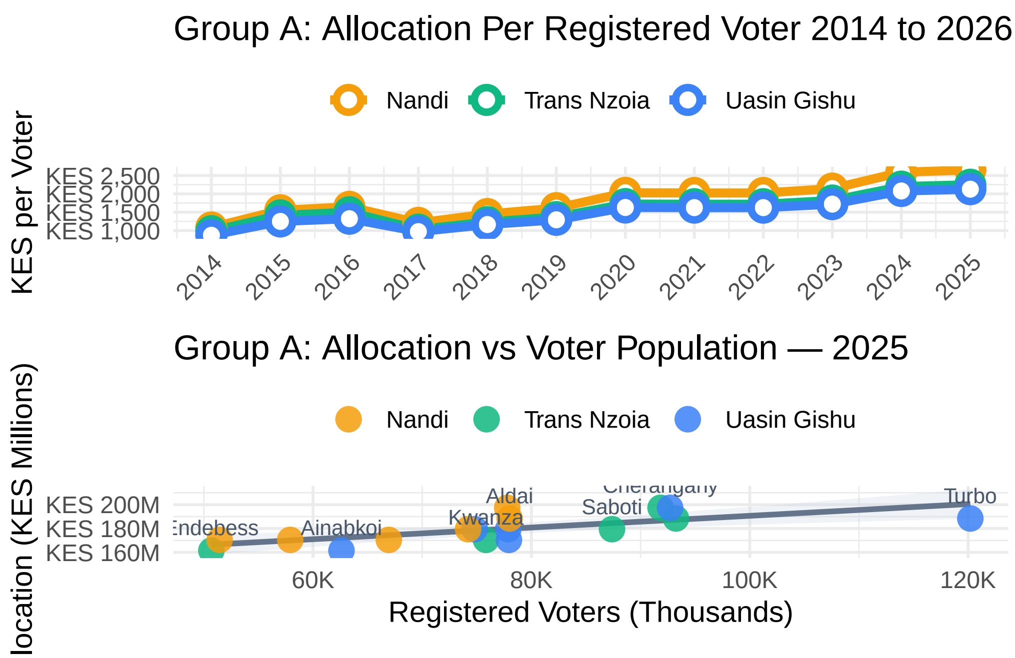

Figure 6: Group A — Per-registered-voter allocation trends (top) and scatter of 2025 allocation vs voter population by constituency (bottom).

4.2.7 National Ranking (2025)

Understanding how Group A counties compare to all 47 Kenyan counties provides vital context for interpreting their allocation levels. A county may appear to be receiving substantial funds in absolute terms, but if it ranks in the bottom quartile nationally, it is in fact among the less well-funded counties relative to its peers. Conversely, a county ranking highly may face greater scrutiny over whether its allocations are being deployed effectively.

The national ranking presented here is based on the **total NGCDF allocation received across all constituencies within each county in 2025**. This means that counties with more constituencies (and therefore more MPs) will tend to rank higher simply by virtue of having more funding streams. To account for this, the ranking table also shows the **average per-constituency allocation**, which enables a fairer comparison of how generously each county's constituencies are funded on average.

The bar chart ranks all 47 counties from bottom to top by 2025 allocation, with Group A counties highlighted in their distinct colors. This makes it immediately clear whether these counties are clustered near the top, middle, or bottom of the national distribution. Counties that are geographically adjacent or economically comparable to the study counties provide useful benchmarks.

```r
rank_A %>%
  mutate(
    county    = fct_reorder(county, total_2025),
    highlight = if_else(county %in% GROUP_A, as.character(county), "Other")) %>%
  ggplot(aes(x = total_2025 / 1e9, y = county, fill = highlight, alpha = highlight)) +
  geom_col(width = 0.8) +
  scale_fill_manual(values = c(COL[GROUP_A], Other = "#cbd5e1")) +
  scale_alpha_manual(values = c(setNames(rep(1, 3), GROUP_A), Other = 0.5)) +
  scale_x_continuous(labels = label_number(suffix = "B", prefix = "KES ")) +
  labs(
    title    = "Group A: All 47 Counties - 2025 NGCDF Ranking",
    subtitle = "Group A counties highlighted",
    x = "Total Allocation (KES Billion)", y = NULL) +
  guides(fill = guide_legend(title = NULL), alpha = "none") +
  theme(legend.position = "top", axis.text.y = element_text(size = 8.5))
```

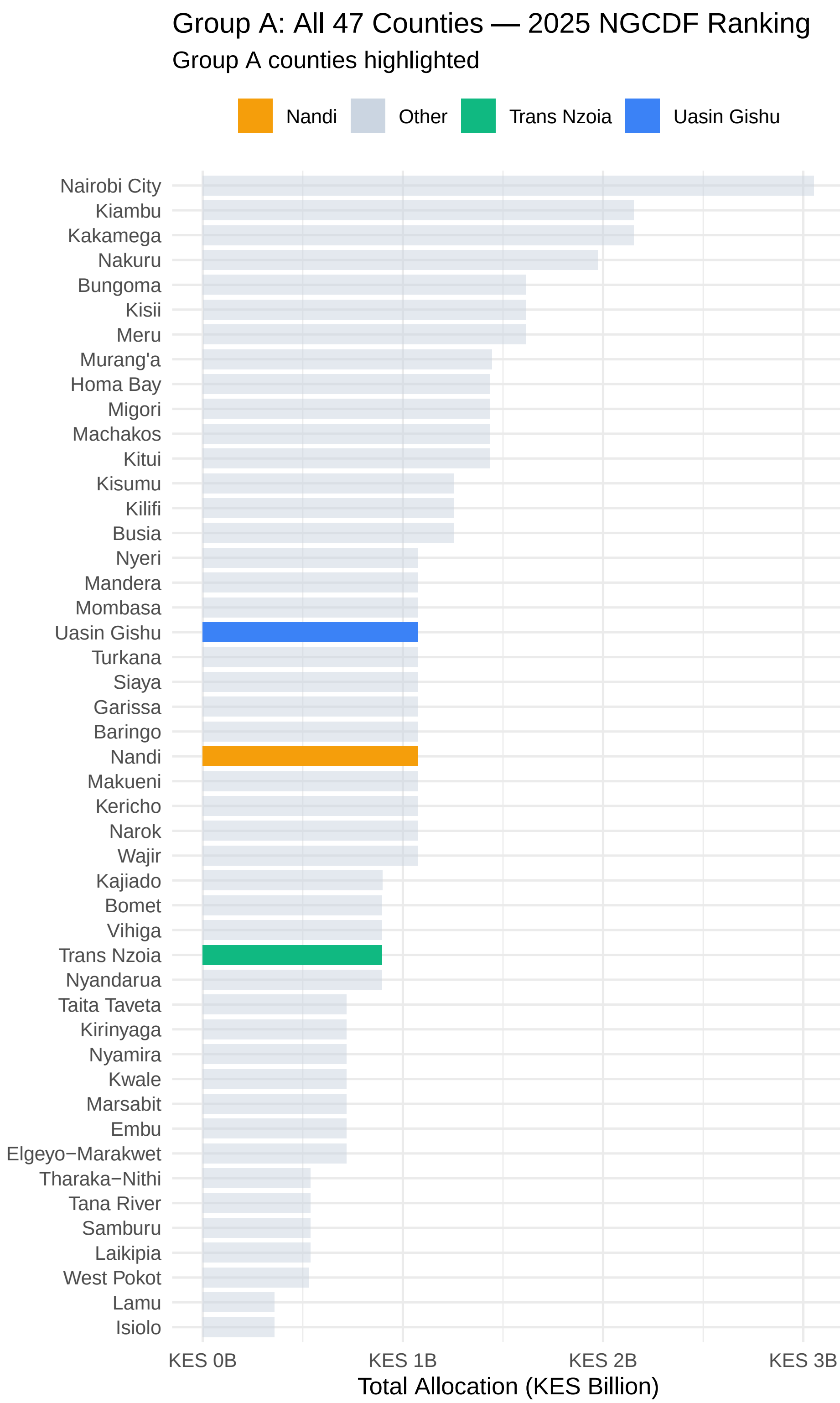

Figure 7: Group A: All 47 counties ranked by 2025 NGCDF allocation; Group A counties highlighted.

Table 6: Group A — National ranking positions for Trans Nzoia, Nandi, and Uasin Gishu (2025)

County	Rank	Total 2025 (KES)	No. Constituencies	Avg per Const. (KES)
Uasin Gishu	23	1,076,651,727	6	179,441,954
Nandi	26	1,076,651,727	6	179,441,954
Trans Nzoia	31	897,209,772	5	179,441,954

```
rank_A %>%
  filter(is_target) %>%
  select(
    County = county, Rank = rank,
    `Total 2025 (KES)` = total_2025,
    `No. Constituencies` = n_const,
    `Avg per Const. (KES)` = avg_per_const) %>%
  mutate(
    `Total 2025 (KES)`     = format(round(`Total 2025 (KES)`), big.mark = ","),
    `Avg per Const. (KES)` = format(round(`Avg per Const. (KES)`), big.mark = ",")) %>%
  kbl(align = c("l", "c", "r", "c", "r")) %>%
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover"))
```

4.2.8 Comparison with Neighboring Counties

Peer comparison with geographically proximate counties is a powerful analytical tool. Counties that share similar demographic profiles, economic structures, and infrastructure conditions should, in theory, receive broadly comparable NGCDF allocations. Where significant differences exist, it raises questions worth examining: Are the disparities driven by differences in the number of constituencies? By different voter registration rates? Or by structural factors in the allocation formula?

For Group A, the selected neighboring counties are: **Elgeyo-Marakwet, Baringo, Kakamega, West Pokot, and Nyandarua**. These counties share borders or are part of the broader Rift Valley and Western corridor. Including them provides a regional lens on whether Trans Nzoia, Nandi, and Uasin Gishu are receiving more, less, or roughly comparable funding relative to their neighbors.

Reading the Comparison Charts

The **left panel** is a snapshot of 2025 allocations, ranking all counties in the peer group from lowest to highest. Target (Group A) counties are shown at full opacity; neighboring counties appear slightly faded. Exact allocation values are labelled at the end of each bar.

The **right panel** shows trends from 2014 to 2026. Target counties are drawn with thick solid lines; neighbors use thinner dashed lines. This makes it straightforward to track whether Group A counties have been consistently above, below, or crossing the trajectories of their neighbors over time. A county that was once below a neighbor but has since overtaken it signals a shift in relative resource allocation one worth understanding in policy terms.

```
SHOW_A <- c(GROUP_A, NEIGHBOURS_A)

nb_yr_A <- df_all %>%
  filter(county %in% SHOW_A) %>%
  pivot_longer(all_of(YEARS), names_to = "year", values_to = "alloc") %>%
  mutate(year = as.integer(year)) %>% filter(!is.na(alloc)) %>% group_by(county, year) %>%
  summarise(total = sum(alloc), n = n(), .groups = "drop")
pN1_A <- nb_yr_A %>%
  filter(year == 2025) %>%
  arrange(desc(total)) %>%
  mutate(county = fct_inorder(county),
         type   = ifelse(county %in% GROUP_A, "Target", "Neighbour")) %>%
  ggplot(aes(x = total / 1e9, y = county, fill = county, alpha = type)) +
  geom_col(width = 0.75, show.legend = FALSE) +
  geom_text(aes(label = paste0(round(total / 1e9, 2), "B")),
            hjust = -0.15, size = 3, colour = "#475569") +
  scale_fill_manual(values = COL) +
  scale_alpha_manual(values = c("Target" = 1, "Neighbour" = 0.55)) +
  scale_x_continuous(
    labels = label_number(suffix = "B", prefix = "KES "),
    expand = expansion(mult = c(0, 0.25))) +
  labs(title = "2025 Allocation: Target vs Neighbours",
       x = "Total (KES Billion)", y = NULL)
pN2_A <- nb_yr_A %>%
  mutate(type = ifelse(county %in% GROUP_A, "Target", "Neighbour")) %>%
  ggplot(aes(x = year, y = total / 1e9, colour = county, group = county,
             linewidth = type, linetype = type)) +
  geom_line() + geom_point(data = ~ filter(.x, type == "Target"), size = 2.5) +
  scale_colour_manual(values = COL) +
  scale_linewidth_manual(values = c("Target" = 2, "Neighbour" = 1), guide = "none") +
  scale_linetype_manual(values = c("Target" = "solid", "Neighbour" = "dashed"), guide = "none
  scale_x_continuous(breaks = c(2014, 2017, 2020, 2023, 2026)) +
  scale_y_continuous(labels = label_number(suffix = "B", prefix = "KES ")) +
  labs(title = "Allocation Trends - Target + Neighbours",
       x = NULL, y = "Total (KES Billion)", colour = NULL) +
  theme(legend.position = "top")
pN1_A | pN2_A
```

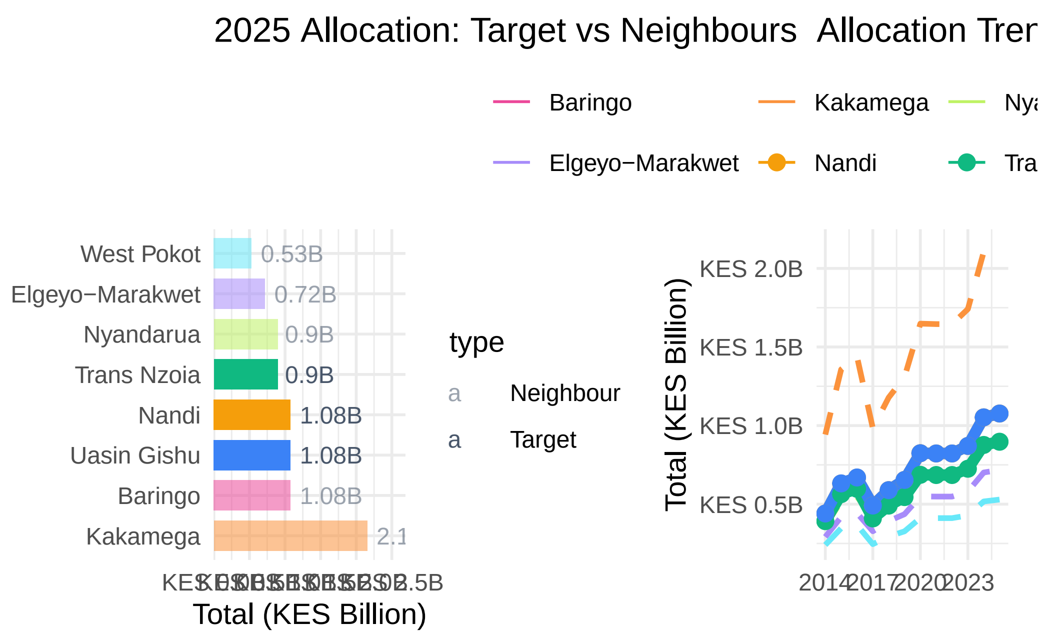

Figure 8: Group A — 2025 allocations (left) and allocation trends 2014–2026 (right) for target and neighbouring counties.

4.2.9 Statistical Tests

The visual and descriptive analysis above tells a compelling story, but statistical testing provides the rigor needed to distinguish genuine differences from random variation. This section presents three types of tests, each addressing a different analytical question.

ANOVA and Kruskal-Wallis: Are County Differences Real?

The **one-way Analysis of Variance (ANOVA)** tests whether the mean 2025 constituency allocation differs significantly across the three counties within Group A. The null hypothesis is that all counties have the same mean — that any observed differences are due to chance. A statistically significant result ($p < 0.05$) means we can reject this and conclude that at least one county's constituencies receive systematically different allocations from the others.

ANOVA assumes that residuals are approximately normally distributed and that variances are similar across groups. With the relatively small number of constituencies in each county, these assumptions may not always hold, so the **Kruskal-Wallis test** is also presented as a non-parametric alternative. It makes no assumption about the distribution of the data and tests whether the distributions of allocations differ across counties. If both tests agree, the findings are more robust.

Paired t-test: Has Funding Changed Year-on-Year?

The **paired t-test (2024 vs 2025)** asks a more targeted question: did constituency allocations change significantly from one year to the next within each county? By treating the 2024 and 2025 allocations of the same constituency as a "pair", this test controls for constituency-level differences and focuses purely on whether there has been a systematic upward or downward shift in funding. A significant result ($p < 0.05$) indicates a genuine year-on-year change beyond what random variation would produce. The mean difference (positive or negative) tells us the average direction and magnitude of that change per constituency.

Pearson Correlation: Does Voter Count Predict Allocation?

The **Pearson correlation coefficient (r)** measures the linear association between the number of registered voters and the 2025 NGCDF allocation at the constituency level. An r close to +1 means that constituencies with more voters tend to receive more funding which would be consistent with a population-weighted allocation formula. An r close to 0 means voter count is a weak predictor of allocation, suggesting other factors dominate. A negative r would be surprising and would warrant investigation.

The strength of the correlation is classified as:

r	Strength
0.0 – 0.29	Weak
0.3 – 0.59	Moderate
0.6 – 1.0	Strong

```
anova_df_A <- df_A %>% filter(!is.na(`2025`)) %>% mutate(county = factor(county))
aov_A <- aov(`2025` ~ county, data = anova_df_A)
kw_A  <- kruskal.test(`2025` ~ county, data = anova_df_A)

aov_sum <- summary(aov_A)[[1]]

tibble(
  Test      = c("One-way ANOVA", "Kruskal-Wallis"),
  Statistic = c(round(aov_sum$`F value`[1], 3), round(kw_A$statistic, 3)),
  df        = c(paste0(aov_sum$Df[1], ", ", aov_sum$Df[2]), kw_A$parameter),
  `p-value` = c(round(aov_sum$`Pr(>F)`[1], 4), round(kw_A$p.value, 4)),
  Conclusion = c(
```

Table 8: Group A: ANOVA and Kruskal-Wallis tests on 2025 constituency allocations

Test	Statistic	df	p-value	Conclusion
One-way ANOVA	0.000	2, 14	1.0000	No significant difference
Kruskal-Wallis	0.003	2	0.9983	No significant difference

Table 9: Group A: Paired t-tests: 2024 vs 2025 allocation per constituency

County	t-statistic	Mean Diff (KES)	p-value	Significance
Trans Nzoia	28.284	4,080,144	0	Significant
Nandi	39.299	4,075,477	0	Significant
Uasin Gishu	34.569	4,075,211	0	Significant

```
    ifelse(aov_sum$`Pr(>F)`[1] < 0.05, "Significant difference", "No significant difference")
    ifelse(kw_A$p.value < 0.05, "Significant difference", "No significant difference")
  )
) %>%
  kbl(align = c("l", "r", "c", "r", "l")) %>%
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover"))
```

```
map_dfr(GROUP_A, function(co) {
  sub <- df_A %>% filter(county == co, !is.na(`2024`), !is.na(`2025`))
  tt  <- t.test(sub$`2025`, sub$`2024`, paired = TRUE)
  tibble(
    County            = co,
    `t-statistic`     = round(tt$statistic, 3),
    `Mean Diff (KES)` = format(round(tt$estimate), big.mark = ","),
    `p-value`         = round(tt$p.value, 4),
    Significance      = ifelse(tt$p.value < 0.05, "Significant  ", "Not Significant"))}) %>%
  kbl(align = c("l", "r", "r", "r", "l")) %>%
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover"))
```

```
map_dfr(GROUP_A, function(co) {
  sub <- df_A %>% filter(county == co, !is.na(`2025`), !is.na(voters), voters > 0)
  ct  <- cor.test(sub$voters, sub$`2025`)
  tibble(
    County    = co,
    `r`       = round(ct$estimate, 4),
    `p-value` = round(ct$p.value, 4),
    Strength  = case_when(
```

Table 10: Group A: Pearson correlation between registered voters and 2025 allocation

County	r	p-value	Strength
Trans Nzoia	0.8930	0.0413	Strong
Nandi	0.8194	0.0460	Strong
Uasin Gishu	0.7241	0.1037	Strong

```
      abs(ct$estimate) >= 0.6 ~ "Strong",
      abs(ct$estimate) >= 0.3 ~ "Moderate",
      TRUE                    ~ "Weak"))}) %>%
  kbl(align = c("l", "r", "r", "l")) %>%
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover"))
```

4.3 Group B: Central: Murang'a, Kiambu & Kirinyaga

The three Central region counties in this group — Murang'a, Kiambu, and Kirinyaga, occupy the fertile slopes and foothills of Mount Kenya, one of Kenya's most agriculturally productive zones. Together they form part of what is historically known as the Central Province, sharing cultural and economic ties while exhibiting distinct development profiles.

Kiambu is the most urbanized of the three, benefiting from its immediate adjacency to Nairobi. Large portions of Kiambu have been absorbed into the Nairobi metropolitan area, making it one of Kenya's fastest-growing counties with a dynamic mix of residential development, manufacturing, and trade. Its proximity to the capital also means it competes with Nairobi's infrastructure for workers and services, creating a unique development context. With 12 constituencies, Kiambu is also the largest county in the group by constituency count, which naturally drives its higher absolute allocation totals.

Murang'a sits to the north of Kiambu and is primarily characterized by smallholder tea and coffee farming, along with significant horticultural production. It has 10 constituencies and has historically been one of the more agricultural and rural counties in the Central belt, though it is increasingly connected to Nairobi's economic sphere.

Kirinyaga is the smallest of the three by constituency count (7), located in the rice-growing Mwea plains and the coffee and tea highlands of the Mount Kenya slopes. Mwea is one of East Africa's most productive irrigated rice-farming areas, and Kirinyaga's agricultural identity strongly shapes its development needs and priorities.

4.3.1 Allocation Trends and Forecasts

As with Group A, the allocation trend chart for Group B reveals the trajectory of NGCDF funding across the three counties from 2014 to 2026, with linear projections extending to 2030. Because Kiambu has significantly more constituencies than Murang'a and Kirinyaga, its county total will naturally be higher — this is expected and not a sign of preferential treatment. The per-voter and per-constituency metrics presented later in this section provide the fairer comparison.

Look for divergences or convergences in the trend lines over time. If Kirinyaga's slope is steeper than Kiambu's, the smaller county may be closing the per-constituency funding gap. Conversely, if the three trends are roughly parallel, it suggests the allocation formula is applying a consistent multiplier across all three counties year on year.

```
trend_B <- cy_B %>% select(county, year, total) %>% mutate(type = "Actual") %>% bind_rows(fc_
ggplot(trend_B, aes(x = year, y = total / 1e9,
                    colour = county,
                    linetype = type,
                    group = interaction(county, type))) +
  geom_vline(xintercept = 2026.5, colour = "#94a3b8", linetype = "dashed", linewidth = 0.6) +
  geom_line(linewidth = 1.5) +
  geom_point(data = ~ filter(.x, type == "Actual"),
             size = 2.5, shape = 21, fill = "white", stroke = 2) +
  geom_point(data = ~ filter(.x, type == "Forecast"),
             size = 3, shape = 17) +
  annotate("text", x = 2027.3, y = max(trend_B$total / 1e9) * 0.55,
           label = "Forecast →", colour = "#6366f1", size = 3.2, fontface = "italic") +
  scale_colour_manual(values = COL) +
  scale_linetype_manual(values = c("Actual" = "solid", "Forecast" = "dashed")) +
  scale_x_continuous(breaks = c(2014, 2017, 2020, 2023, 2026, 2027, 2030)) +
  scale_y_continuous(labels = label_number(suffix = "B", prefix = "KES ")) +
  labs(
    title    = "Group B: NGCDF County Totals - Actuals + Forecast 2027-2030",
    subtitle = "Linear trend fitted on 2020-2026 trajectory",
    x = NULL, y = "Total (KES Billion)", colour = NULL, linetype = NULL) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1), legend.position = "top")
```

Table 11: Group B — Projected NGCDF allocations 2027–2030 (KES, rounded)

	Projected Allocation (KES)		
Year	Kiambu	Kirinyaga	Murang'a
2027	2,336,613,847	780,171,305	1,572,755,744
2028	2,450,699,990	818,406,761	1,651,221,092
2029	2,564,786,134	856,642,217	1,729,686,440
2030	2,678,872,277	894,877,674	1,808,151,788

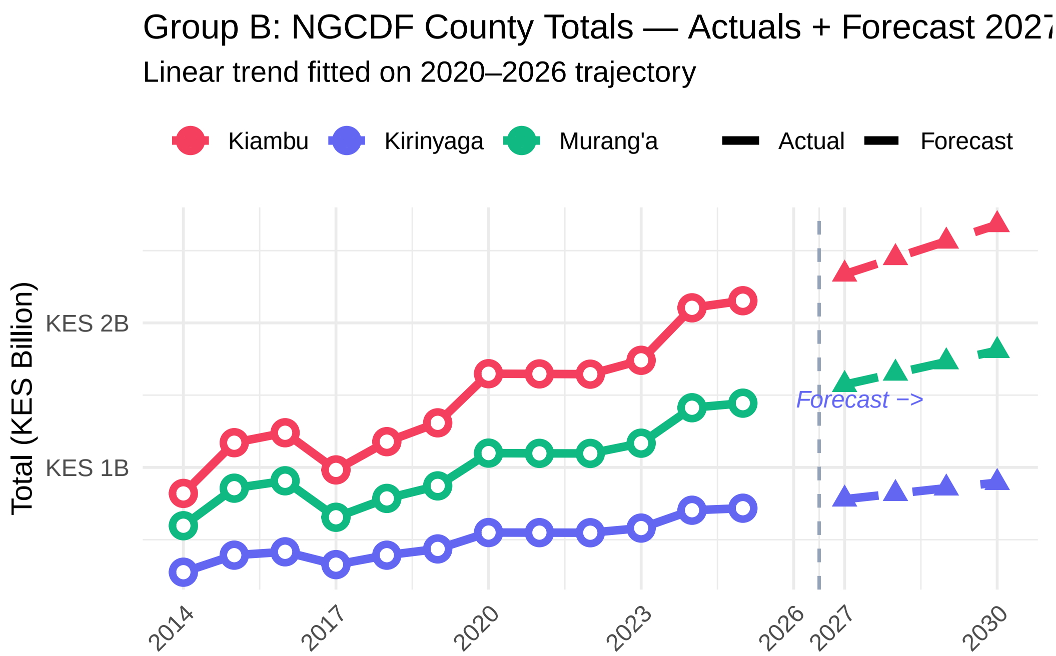

Figure 9: Group B: Annual NGCDF county totals (actuals 2014–2026) and linear forecast 2027–2030.

```
fc_B %>%
  mutate(total = round(total)) %>%
  pivot_wider(names_from = county, values_from = total, id_cols = year) %>%
  rename(Year = year) %>%
  mutate(across(-Year, ~ format(.x, big.mark = ","))) %>%
  kbl(align = c("l", rep("r", 3))) %>%
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover", "condensed")) %
  add_header_above(c(" " = 1, "Projected Allocation (KES)" = 3))
```

4.3.2 Constituency Performance Tiers (2025)

As with Group A, the tier analysis for Group B classifies each constituency into High, Mid, or Low tiers based on where its 2025 allocation falls within the county's own distribution. This is a purely relative classification it identifies which constituencies within each county are relatively well-funded and which are relatively under-funded compared to their county peers.

For Kiambu in particular, the tier analysis is especially informative because its 12 constituencies create a wider spread of possible allocation outcomes than in smaller counties. A significant gap between the High and Low tier constituencies in Kiambu could reflect the urban-rural divide within the county. Urban constituencies near Nairobi may attract different types of projects with different cost profiles compared to constituencies in the rural north of the county.

```r
tier_B %>%
  ggplot(aes(x = `2025` / 1e6, y = reorder(constituency, `2025`), fill = tier)) +
  geom_col(width = 0.75) +
  geom_text(aes(label = tier), hjust = -0.2, size = 2.8, colour = "#475569") +
  scale_fill_manual(values = c("High" = "#10b981", "Mid" = "#f59e0b", "Low" = "#ef4444")) +
  scale_x_continuous(
    labels = label_number(suffix = "M", prefix = "KES "),
    expand = expansion(mult = c(0, 0.25))) +
  facet_wrap(~ county, scales = "free_y", ncol = 3) +
  labs(
    title    = "Group B: Constituency Performance Tiers - 2025",
    subtitle = "High = top third · Mid = middle third · Low = bottom third",
    x = "Allocation (KES Millions)", y = NULL, fill = "Tier") +
  theme(legend.position = "top")
```

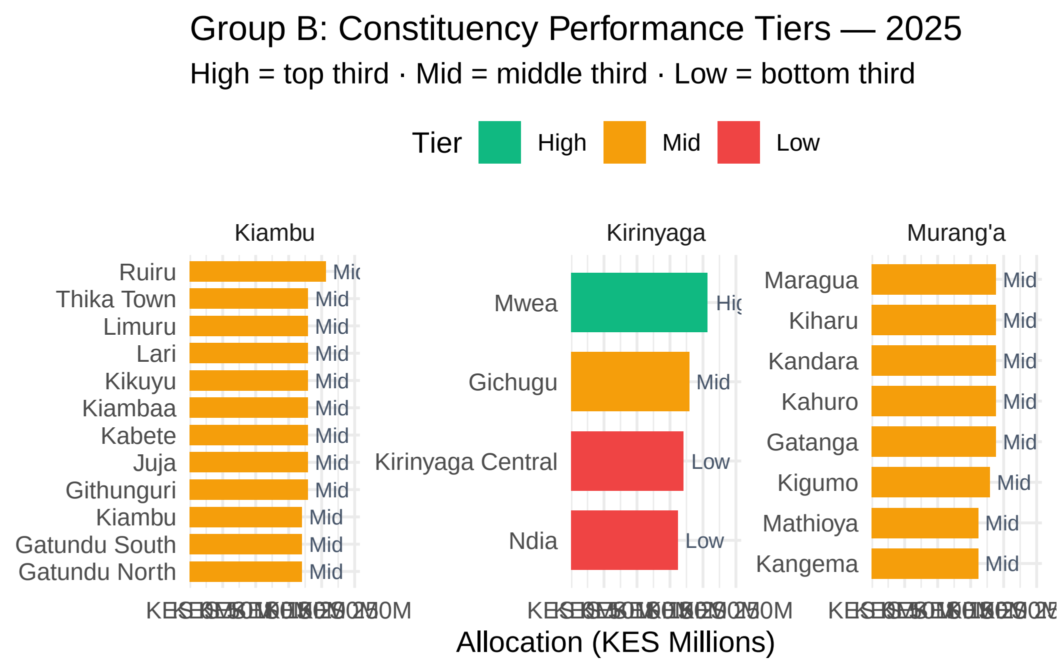

Figure 10: Group B — Constituency-level 2025 allocations by performance tier.

4.3.3 Equity and Intra-County Distribution (Gini)

The Central region counties present a particularly interesting equity story. Kiambu's large number of constituencies and its mix of urban and rural areas create conditions for potentially high within-county inequality. If urban constituencies near Nairobi consistently receive more (in absolute terms) than rural ones further north, the Gini coefficient will reflect this structural disparity.

Murang'a and Kirinyaga, being smaller and more homogeneous in their agricultural character, may show lower Gini coefficients, though this should not be assumed without examining the data. A persistently high or rising Gini in any of these counties would warrant policy dialogue between the NGCDF Board and the respective CDFCs to understand the drivers and assess whether formula adjustments are needed.

As before, both the Gini coefficient (top panel) and the Coefficient of Variation (bottom panel) are shown. Cross-referencing the two provides a more reliable signal than either metric alone.

```
pE1_B <- gini_B %>%
  ggplot(aes(x = year, y = gini, colour = county, group = county)) +
  geom_line(linewidth = 1.6) +
  geom_point(size = 3, shape = 21, fill = "white", stroke = 2) +
```

```r
  scale_colour_manual(values = COL) +
  scale_x_continuous(breaks = 2014:2026) +
  scale_y_continuous(limits = c(0, NA)) +
  labs(
    title    = "Group B: Intra-County Gini - 2014 to 2026",
    subtitle = "0 = perfectly equal · Higher = more unequal",
    x = NULL, y = "Gini Coefficient", colour = NULL
  ) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1), legend.position = "top")

pE2_B <- gini_B %>%
  ggplot(aes(x = year, y = cv, colour = county, group = county)) +
  geom_line(linewidth = 1.6) +
  geom_point(size = 3, shape = 21, fill = "white", stroke = 2) +
  scale_colour_manual(values = COL) +
  scale_x_continuous(breaks = 2014:2026) +
  scale_y_continuous(labels = label_percent(scale = 1)) +
  labs(
    title = "Group B: Coefficient of Variation (CV)",
    x = NULL, y = "CV (%)", colour = NULL) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1), legend.position = "none")
pE1_B / pE2_B
```

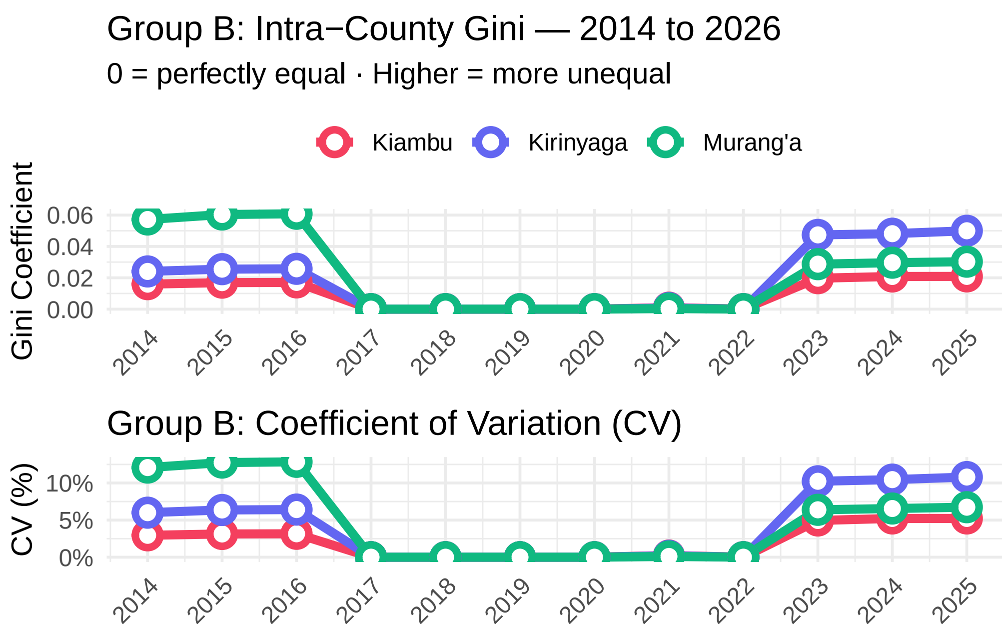

Figure 11: Group B: Gini coefficient and Coefficient of Variation, 2014–2026.

4.3.4 CPI-Adjusted (Real) Allocations

Central Kenya, and Kiambu in particular, is subject to a distinct inflationary environment because of its urban proximity to Nairobi. Construction costs, land prices, and labor rates in Kiambu are significantly higher than in most other Kenyan counties. This means that even if Kiambu's CPI-adjusted allocation looks comparable to rural counties, the county's *effective* purchasing power for development projects may be lower because the prices it faces are above the national average CPI.

This is an important caveat to keep in mind when interpreting the CPI-adjusted figures: the national CPI used in this analysis is a single Kenya-wide index and does not capture county-level cost differences. Kiambu's real allocation may therefore be overstated relative to what it can actually procure locally. For Murang'a and Kirinyaga, the national CPI deflator is likely a more accurate approximation of their cost environment, making their real allocation figures more directly comparable.

```
cpi_B <- cy_B %>%
  select(county, year, total, real_total) %>%
  mutate(
    nominal_idx = total      / total[year == 2014]      * 100,
    real_idx    = real_total / real_total[year == 2014] * 100,
```

```
    .by = county)
cpi_B %>%
  pivot_longer(c(nominal_idx, real_idx), names_to = "type", values_to = "idx") %>%
  mutate(type = recode(type,
    "nominal_idx" = "Nominal (current KES)",
    "real_idx"    = "Real (2014 KES)")) %>%
  ggplot(aes(x = year, y = idx, colour = county, linetype = type)) +
  geom_hline(yintercept = 100, colour = "#94a3b8", linetype = "dotted") +
  geom_line(linewidth = 1.4) +
  geom_point(size = 2) +
  scale_colour_manual(values = COL) +
  scale_linetype_manual(values = c("Nominal (current KES)" = "solid", "Real (2014 KES)" = "da
  scale_x_continuous(breaks = c(2014, 2017, 2020, 2023, 2026)) +
  facet_wrap(~ county, ncol = 3) +
  labs(
    title    = "Group B: Nominal vs CPI-Adjusted Allocation Index",
    subtitle = "Base 2014 = 100",
    x = NULL, y = "Index (2014 = 100)", colour = NULL, linetype = NULL) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1), legend.position = "top")
```

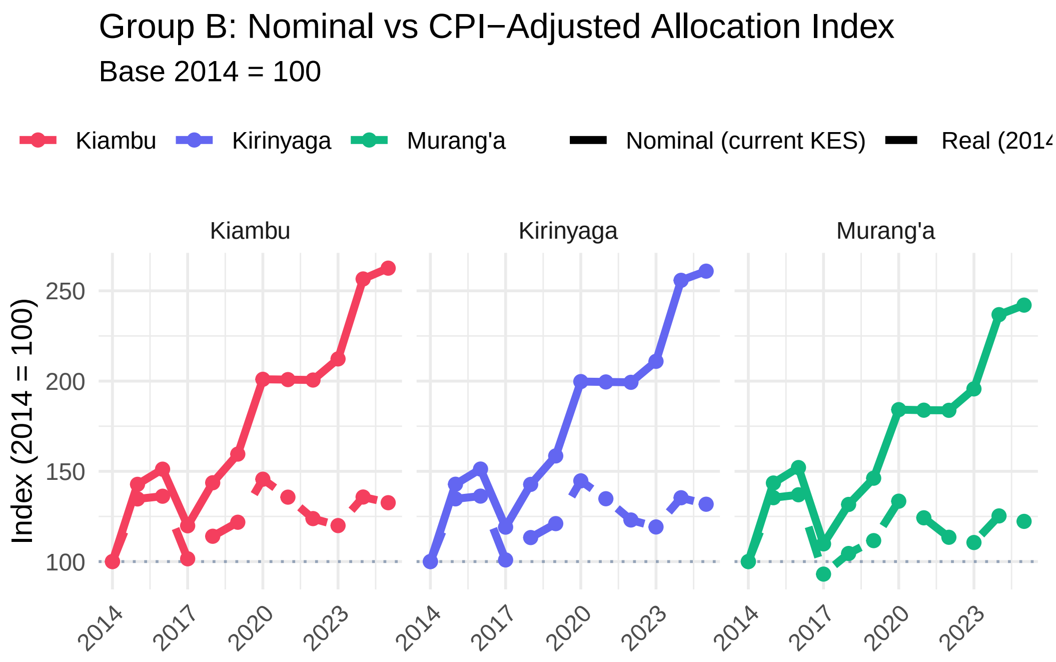

Figure 12: Group B: Nominal vs CPI-adjusted allocation index, 2014 = 100.

4.3.5 Per-Voter Allocation and Voter Context

For the Central region counties, per-voter analysis is particularly revealing because voter registration rates and constituency sizes vary quite markedly. Kiambu has rapidly growing voter rolls driven by urbanization and population in-migration from Nairobi's expanding footprint, which could suppress per-voter allocations even as absolute totals grow. Kirinyaga, with fewer and generally smaller constituencies, may show higher per-voter allocations for some of its constituencies despite receiving a smaller county total.

The scatter plot for 2025 is therefore especially important for this group as it allows a direct visual test of whether the NGCDF allocation formula is compensating (however imperfectly) for constituency size differences, or whether some constituencies are systematically over or under-funded relative to their voter populations.

```
pV1_B <- cy_B %>%
  ggplot(aes(x = year, y = per_voter, colour = county, group = county)) +
  geom_line(linewidth = 1.6) +
  geom_point(size = 3, shape = 21, fill = "white", stroke = 2) +
  scale_colour_manual(values = COL) +
  scale_x_continuous(breaks = 2014:2026) +
  scale_y_continuous(labels = label_number(big.mark = ",", prefix = "KES ")) +
  labs(
    title = "Group B: Allocation Per Registered Voter 2014 to 2026",
    x = NULL, y = "KES per Voter", colour = NULL
  ) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1), legend.position = "top")

pV2_B <- df_B %>%
  filter(!is.na(`2025`), !is.na(voters), voters > 0) %>%
  ggplot(aes(x = voters / 1000, y = `2025` / 1e6, colour = county, label = constituency)) +
  geom_smooth(aes(group = 1), method = "lm", colour = "#64748b",
              se = TRUE, fill = "#cbd5e1", alpha = 0.3, linewidth = 1) +
  geom_point(size = 4, alpha = 0.85) +
  geom_text(size = 2.8, colour = "#475569", vjust = -1, check_overlap = TRUE) +
  scale_colour_manual(values = COL) +
  scale_x_continuous(labels = label_number(suffix = "K")) +
  scale_y_continuous(labels = label_number(suffix = "M", prefix = "KES ")) +
  labs(
    title = "Group B: Allocation vs Voter Population - 2025",
    x = "Registered Voters (Thousands)", y = "Allocation (KES Millions)", colour = NULL) +
  theme(legend.position = "top")
pV1_B / pV2_B
```

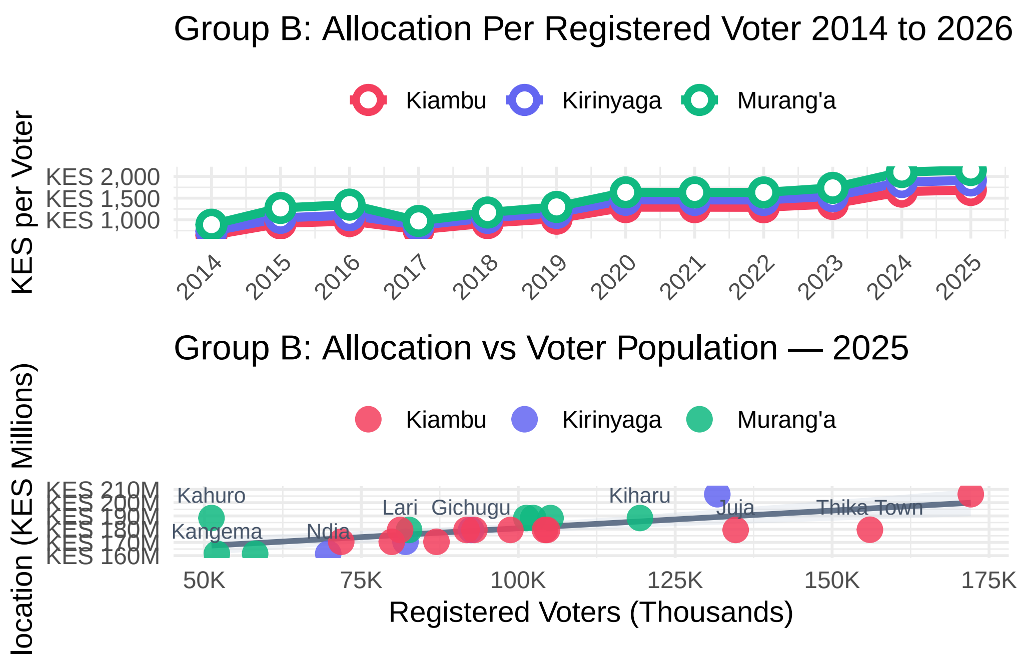

Figure 13: Group B: Per-registered-voter allocation trends and scatter of 2025 allocation vs voter population.

4.3.6 National Ranking (2025)

In the national context, the Central region counties occupy a distinctive position. Kiambu, with its 12 constituencies and high urbanization, would be expected to rank among the top counties by absolute allocation total. Murang'a and Kirinyaga, being smaller, will rank lower but comparing their per-constituency averages to counties of similar size nationally is the more meaningful benchmark.

The ranking chart places all 47 counties on a single canvas, with Group B counties highlighted. This makes it straightforward to see whether Central region counties are broadly in line with, above, or below counties of comparable size and economic profile. Counties that rank unexpectedly low (given their size) or high (relative to their constituency count) may reflect structural features of the allocation formula worth examining.

```
rank_B %>%
  mutate(
    county    = fct_reorder(county, total_2025),
    highlight = if_else(county %in% GROUP_B, as.character(county), "Other")
  ) %>%
  ggplot(aes(x = total_2025 / 1e9, y = county, fill = highlight, alpha = highlight)) +
```

```r
  geom_col(width = 0.8) +
  scale_fill_manual(values = c(COL[GROUP_B], Other = "#cbd5e1")) +
  scale_alpha_manual(values = c(setNames(rep(1, 3), GROUP_B), Other = 0.5)) +
  scale_x_continuous(labels = label_number(suffix = "B", prefix = "KES ")) +
  labs(
    title    = "Group B: All 47 Counties - 2025 NGCDF Ranking",
    subtitle = "Group B counties highlighted",
    x = "Total Allocation (KES Billion)", y = NULL) +
  guides(fill = guide_legend(title = NULL), alpha = "none") +
  theme(legend.position = "top", axis.text.y = element_text(size = 8.5))
```

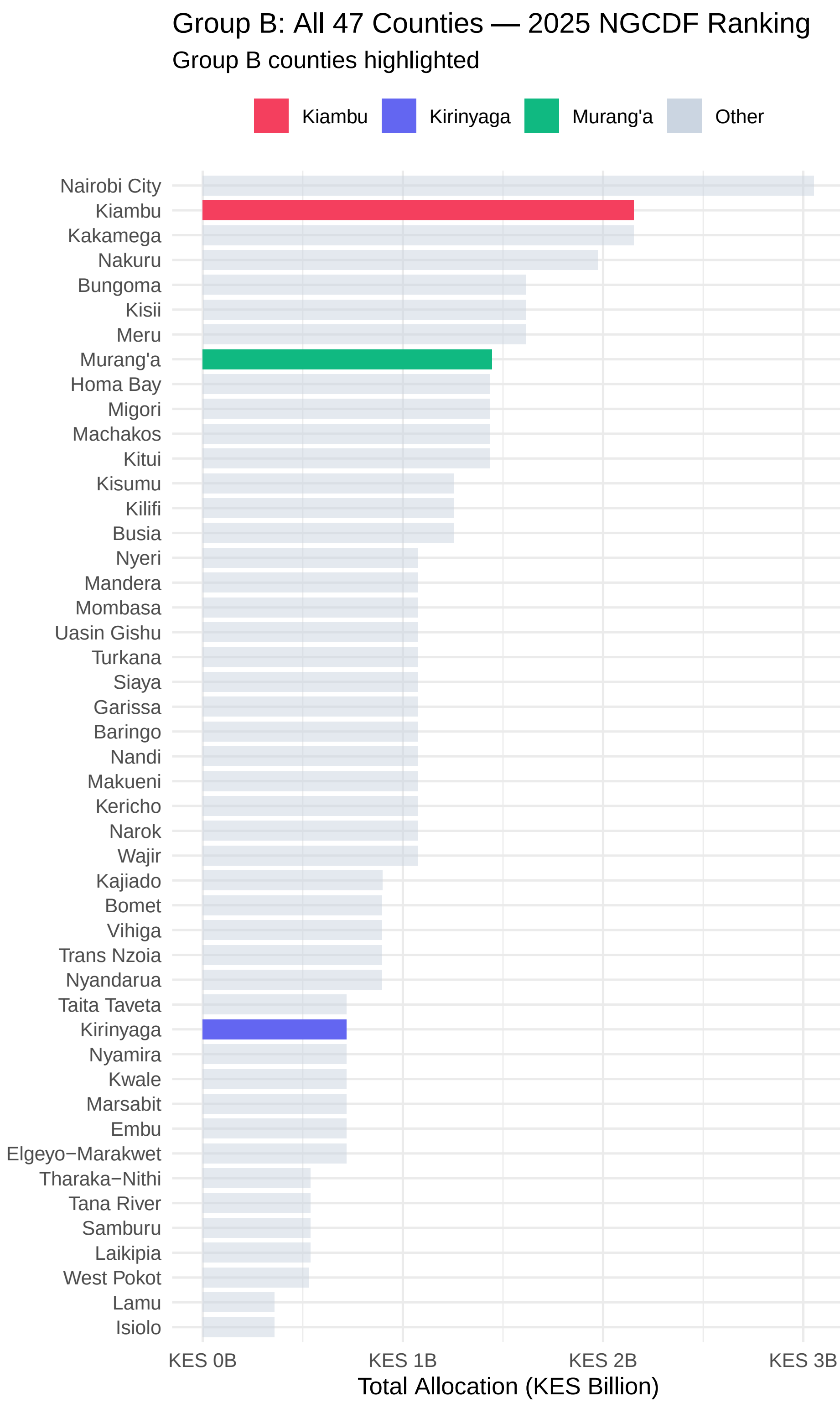

Figure 14: Group B: All 47 counties ranked by 2025 NGCDF allocation; Group B counties highlighted.

Table 12: Group B: National ranking positions for Murang'a, Kiambu, and Kirinyaga (2025)

County	Rank	Total 2025 (KES)	No. Constituencies	Avg per Const. (KES)
Kiambu	2	2,153,440,454	12	179,453,371
Murang'a	8	1,444,507,734	8	180,563,467
Kirinyaga	35	717,779,318	4	179,444,829

```r
rank_B %>%
  filter(is_target) %>%
  select(
    County = county, Rank = rank,
    `Total 2025 (KES)` = total_2025,
    `No. Constituencies` = n_const,
    `Avg per Const. (KES)` = avg_per_const) %>%
  mutate(
    `Total 2025 (KES)`     = format(round(`Total 2025 (KES)`), big.mark = ","),
    `Avg per Const. (KES)` = format(round(`Avg per Const. (KES)`), big.mark = ",")) %>%
  kbl(align = c("l", "c", "r", "c", "r")) %>%
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover"))
```

4.3.7 Comparison with Neighboring Counties

For the Central counties, the selected neighbors are: **Nyandarua, Embu, Machakos, Nairobi, and Nyeri**. This peer group is deliberately diverse as it includes Nairobi, a clear outlier in terms of economic size and constituency count, as well as smaller counties like Embu and Nyandarua that are more comparable to Kirinyaga and Murang'a respectively.

Including Nairobi is instructive precisely because of its outlier status: it illustrates how the NGCDF formula functions at the extreme end of constituency scale, and it contextualizes whether the Central counties are securing allocations proportionate to their relative development needs and constituency sizes. Nyeri provides a good Mount Kenya region benchmark, sharing similar agricultural and demographic characteristics with Murang'a and Kirinyaga.

As before, target counties are shown with solid, bold lines and neighboring counties with dashed, lighter lines in the trend chart, making it straightforward to track relative trajectories over the full 2014–2026 period.

```r
SHOW_B <- c(GROUP_B, NEIGHBOURS_B)

nb_yr_B <- df_all %>%
  filter(county %in% SHOW_B) %>%
```

```
  pivot_longer(all_of(YEARS), names_to = "year", values_to = "alloc") %>%
  mutate(year = as.integer(year)) %>%
  filter(!is.na(alloc)) %>%
  group_by(county, year) %>%
  summarise(total = sum(alloc), n = n(), .groups = "drop")
pN1_B <- nb_yr_B %>%
  filter(year == 2025) %>%
  arrange(desc(total)) %>%
  mutate(county = fct_inorder(county),
         type   = ifelse(county %in% GROUP_B, "Target", "Neighbour")) %>%
  ggplot(aes(x = total / 1e9, y = county, fill = county, alpha = type)) +
  geom_col(width = 0.75, show.legend = FALSE) +
  geom_text(aes(label = paste0(round(total / 1e9, 2), "B")),
            hjust = -0.15, size = 3, colour = "#475569") +
  scale_fill_manual(values = COL) +
  scale_alpha_manual(values = c("Target" = 1, "Neighbour" = 0.55)) +
  scale_x_continuous(
    labels = label_number(suffix = "B", prefix = "KES "),
    expand = expansion(mult = c(0, 0.25))) +
  labs(title = "2025 Allocation: Target vs Neighbours",
       x = "Total (KES Billion)", y = NULL)
pN2_B <- nb_yr_B %>%
  mutate(type = ifelse(county %in% GROUP_B, "Target", "Neighbour")) %>%
  ggplot(aes(x = year, y = total / 1e9, colour = county, group = county,
             linewidth = type, linetype = type)) +
  geom_line() +
  geom_point(data = ~ filter(.x, type == "Target"), size = 2.5) +
  scale_colour_manual(values = COL) +
  scale_linewidth_manual(values = c("Target" = 2, "Neighbour" = 1), guide = "none") +
  scale_linetype_manual(values = c("Target" = "solid", "Neighbour" = "dashed"), guide = "non
  scale_x_continuous(breaks = c(2014, 2017, 2020, 2023, 2026)) +
  scale_y_continuous(labels = label_number(suffix = "B", prefix = "KES ")) +
  labs(title = "Allocation Trends - Target + Neighbours",
       x = NULL, y = "Total (KES Billion)", colour = NULL) +
  theme(legend.position = "top")
pN1_B | pN2_B
```

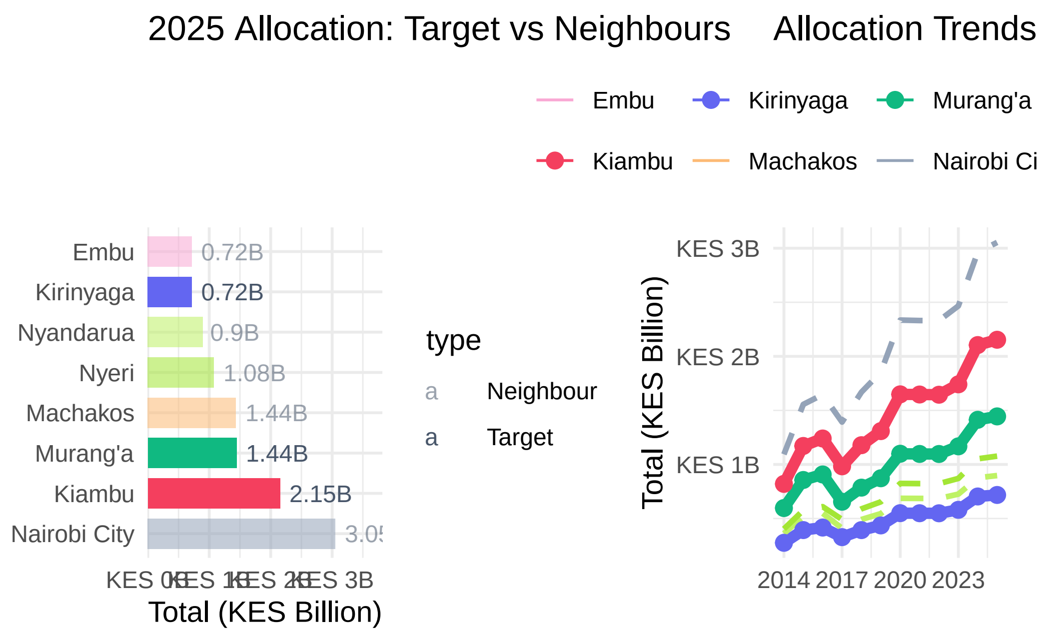

Figure 15: Group B: 2025 allocations and allocation trends for target and neighbouring counties.

4.3.8 Statistical Tests

The same battery of statistical tests applied to Group A is now applied to the Central counties. For Group B, the results may differ in important ways from Group A because the constituency landscape is different: Kiambu's 12 constituencies create more data points for within-county analysis, while Kirinyaga's 7 constituencies mean tests for that county have lower statistical power (a smaller sample is less able to detect true differences).

Keep this in mind when interpreting p-values: a non-significant result in Kirinyaga should not be taken as evidence of equality, but rather as a consequence of limited data. The Kruskal-Wallis test is particularly valuable here as it is robust to unequal sample sizes across groups.

The correlation between voter count and allocation is also worth monitoring closely for Kiambu, given the county's rapidly changing voter landscape. If the correlation is weak or negative for Kiambu, it would suggest that the formula is not fully capturing constituency population dynamics in this urbanizing environment.

```r
anova_df_B <- df_B %>% filter(!is.na(`2025`)) %>% mutate(county = factor(county))
aov_B <- aov(`2025` ~ county, data = anova_df_B)
kw_B  <- kruskal.test(`2025` ~ county, data = anova_df_B)
```

Table 13: Group B — ANOVA and Kruskal-Wallis tests on 2025 constituency allocations

Test	Statistic	df	p-value	Conclusion
One-way ANOVA	0.022	2, 21	0.9781	No significant difference
Kruskal-Wallis	1.043	2	0.5938	No significant difference

Table 14: Group B: Paired t-tests: 2024 vs 2025 allocation per constituency

County	t-statistic	Mean Diff (KES)	p-value	Significance
Murang'a	15.900	3,932,145	0.0000	Significant
Kiambu	65.401	4,091,561	0.0000	Significant
Kirinyaga	6.548	3,492,869	0.0072	Significant

```r
aov_sum_B <- summary(aov_B)[[1]]

tibble(
  Test      = c("One-way ANOVA", "Kruskal-Wallis"),
  Statistic = c(round(aov_sum_B$`F value`[1], 3), round(kw_B$statistic, 3)),
  df        = c(paste0(aov_sum_B$Df[1], ", ", aov_sum_B$Df[2]), kw_B$parameter),
  `p-value` = c(round(aov_sum_B$`Pr(>F)`[1], 4), round(kw_B$p.value, 4)),
  Conclusion = c(
    ifelse(aov_sum_B$`Pr(>F)`[1] < 0.05, "Significant difference", "No significant difference
    ifelse(kw_B$p.value < 0.05, "Significant difference", "No significant difference")
  )
) %>%
  kbl(align = c("l", "r", "c", "r", "l")) %>%
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover"))
```

```r
map_dfr(GROUP_B, function(co) {
  sub <- df_B %>% filter(county == co, !is.na(`2024`), !is.na(`2025`))
  tt  <- t.test(sub$`2025`, sub$`2024`, paired = TRUE)
  tibble(
    County            = co,
    `t-statistic`     = round(tt$statistic, 3),
    `Mean Diff (KES)` = format(round(tt$estimate), big.mark = ","),
    `p-value`         = round(tt$p.value, 4),
    Significance      = ifelse(tt$p.value < 0.05, "Significant  ", "Not Significant"))}) %>%
  kbl(align = c("l", "r", "r", "r", "l")) %>%
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover"))
```

Table 15: Group B: Pearson correlation between registered voters and 2025 allocation

County	r	p-value	Strength
Murang'a	0.6827	0.0621	Strong
Kiambu	0.7681	0.0035	Strong
Kirinyaga	0.9993	0.0007	Strong

```r
map_dfr(GROUP_B, function(co) {
  sub <- df_B %>% filter(county == co, !is.na(`2025`), !is.na(voters), voters > 0)
  ct  <- cor.test(sub$voters, sub$`2025`)
  tibble(
    County    = co,
    `r`       = round(ct$estimate, 4),
    `p-value` = round(ct$p.value, 4),
    Strength  = case_when(
      abs(ct$estimate) >= 0.6 ~ "Strong",
      abs(ct$estimate) >= 0.3 ~ "Moderate",
      TRUE                    ~ "Weak"))}) %>%
  kbl(align = c("l", "r", "r", "l")) %>%
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover"))
```

4.4 Cross-Group Comparison: All Six Counties

Having examined each regional group in detail, this section brings all six counties together for a direct cross-group comparison. This is where the analysis moves from describing individual county trajectories to identifying broader patterns about how Rift Valley and Central region counties compare in terms of NGCDF allocations, growth rates, and equity.

The two groups occupy different places in Kenya's development geography. The Rift Valley counties (Group A) are predominantly agricultural, with significant food-production significance at the national level, and are served by a more rural constituency structure. The Central counties (Group B) are more economically diverse, with stronger links to Nairobi's economic core and higher urbanization rates in Kiambu.

Despite these differences, both groups receive NGCDF funds through the same national formula, making cross-group comparison both legitimate and insightful. If one group is consistently receiving more per constituency or per voter, it raises questions about whether the formula is working equitably across Kenya's diverse county contexts.

4.4.1 Allocation Trends 2014–2026

The combined trend chart places all six counties on a single canvas. Solid lines represent Rift Valley (Group A) counties; dashed lines represent Central (Group B) counties. This visual convention makes it easy to see whether one group as a whole has been growing faster than the other, or whether the trajectories have been broadly similar.

Pay attention to any “crossovers” points where a Group A county’s line intersects with a Group B county’s line. These signal a shift in relative allocation standing and may reflect changes in the voter registration base, constituency redistricting, or shifts in the national formula.

Also note the **spread within each group**: if Group B counties are more dispersed (wider gap between the highest and lowest lines in the group) than Group A, it suggests greater within-group heterogeneity in Central Kenya’s NGCDF receipts largely a reflection of Kiambu’s dominant size relative to Murang’a and Kirinyaga.

```r
cy_6 %>%
  ggplot(aes(x = year, y = total / 1e9, colour = county, group = county,
             linetype = ifelse(county %in% GROUP_A, "Group A", "Group B"))) +
  geom_line(linewidth = 1.6) +
  geom_point(size = 2.5, shape = 21, fill = "white", stroke = 2) +
  scale_colour_manual(values = COL) +
  scale_linetype_manual(values = c("Group A" = "solid", "Group B" = "dashed"),
                        name = "Group") +
  scale_x_continuous(breaks = c(2014, 2017, 2020, 2023, 2026)) +
  scale_y_continuous(labels = label_number(suffix = "B", prefix = "KES ")) +
  labs(
    title    = "All 6 Target Counties - Allocation Trends 2014-2026",
    subtitle = "Solid = Rift Valley (Group A) · Dashed = Central (Group B)",
    x = NULL, y = "Total (KES Billion)", colour = NULL
  ) +
  theme(legend.position = "top")
```

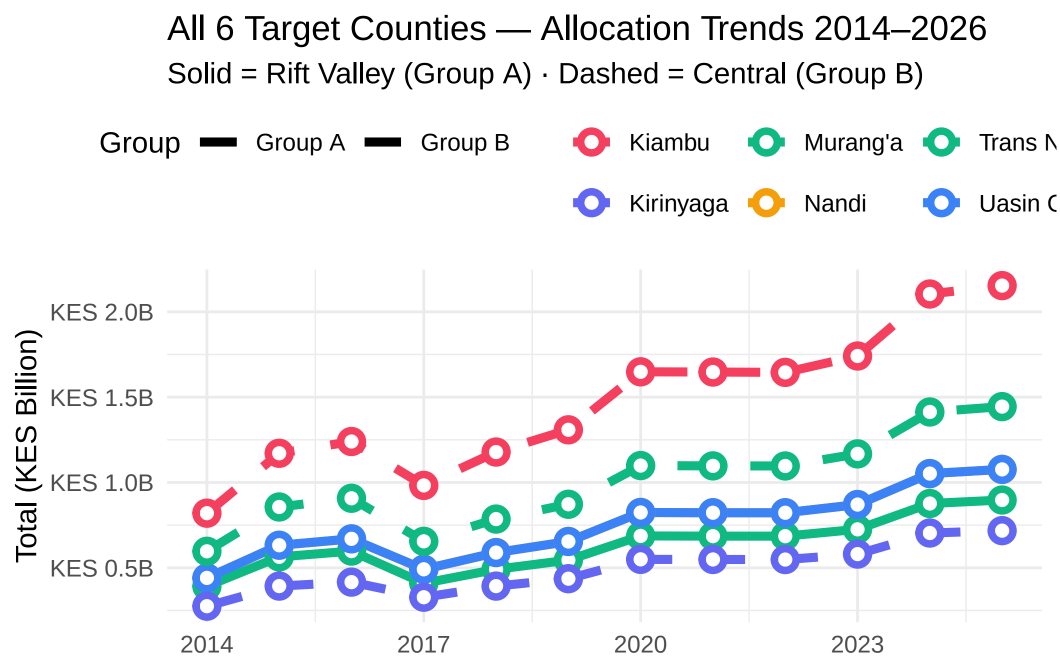

Figure 16: All six target counties, NGCDF allocation trends 2014–2026. Solid lines = Group A (Rift Valley); dashed lines = Group B (Central).

4.4.2 2025 Allocation Side-by-Side

The 2025 allocation comparison, faceted by group, provides the clearest snapshot of where each county stands at the most recent point in the data series. Within each group, counties are sorted from lowest to highest allocation, with exact values labelled.

A few things are worth examining closely here. First, how does the within-group spread compare between Group A and Group B? A tighter spread (counties closer together in allocation level) suggests more uniform development investment within the group. A wider spread, especially if one county significantly dominates, points to structural differences (typically constituency count) that absolute totals cannot overcome.

Second, how do the highest-ranked Group A county and the lowest-ranked Group B county compare? If there is significant overlap between the groups' allocation ranges, it suggests the two groups are broadly in the same tier of national NGCDF receipts. If Group B counties uniformly exceed Group A, it signals a structural advantage, most likely Kiambu's out-sized constituency count.

The scorecard table that follows this chart brings together the key metrics, 2014 and 2025 allocations, nominal growth, national rank, and Gini coefficient into a single consolidated view for all six counties, enabling a comprehensive side-by-side assessment.

```r
cy_6 %>%
  filter(year == 2025) %>%
  mutate(
    county = fct_reorder(county, total),
    group  = ifelse(county %in% GROUP_A, "Rift Valley (Group A)", "Central (Group B)")
  ) %>%
  ggplot(aes(x = total / 1e9, y = county, fill = county)) +
  geom_col(width = 0.75) +
  geom_text(aes(label = paste0("KES ", round(total / 1e9, 2), "B")),
            hjust = -0.1, size = 3.2, colour = "#475569") +
  scale_fill_manual(values = COL) +
  scale_x_continuous(
    labels = label_number(suffix = "B", prefix = "KES "),
    expand = expansion(mult = c(0, 0.3))
  ) +
  facet_wrap(~ group, scales = "free_y", ncol = 1) +
  labs(
    title = "2025 Allocation - All 6 Counties by Group",
    x = "Total Allocation (KES Billion)", y = NULL
  ) +
  theme(legend.position = "none")
```

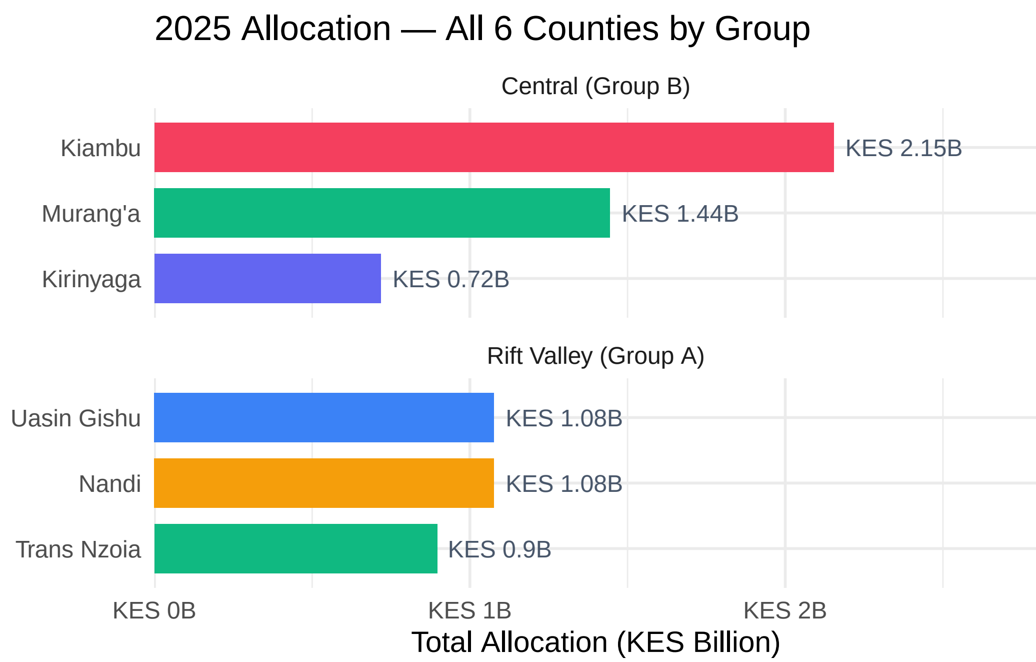

Figure 17: All six target counties — 2025 NGCDF allocations faceted by regional group.

4.4.3 Summary Scorecard

The scorecard below is the single most compact summary of the entire analysis. Each row represents one county, and together the columns capture the most important dimensions of NGCDF performance:

- **2014 and 2025 Allocations (KES):** The starting and ending points of the nominal allocation series, showing the absolute scale of funding each county receives.
- **Growth (%):** The nominal percentage increase from 2014 to 2025. This is the headline growth figure, but remember that it includes the effect of inflation — compare with the CPI-adjusted analysis in earlier sections for a clearer picture of real growth.
- **National Rank (2025):** How the county sits within the full landscape of all 47 Kenyan counties by total NGCDF allocation. A lower rank number means more funding.
- **Gini (2025):** The within-county inequality of constituency allocations in the most recent year. Lower values indicate a more equitable distribution; higher values indicate greater concentration among a subset of constituencies.

Reading across any single row gives a quick profile of that county: how big is its NGCDF envelope, how fast has it grown, where does it stand nationally, and how equitably are its constituencies served?

```r
## Combine ranking data
rank_all <- bind_rows(rank_A, rank_B) %>% distinct(county, .keep_all = TRUE)
gini_all <- bind_rows(gini_A, gini_B)

scorecard <- cy_6 %>%
  filter(year %in% c(2014, 2025)) %>%
  select(county, year, total) %>%
  pivot_wider(names_from = year, values_from = total, names_prefix = "y") %>%
  mutate(growth_pct = round((y2025 / y2014 - 1) * 100, 1)) %>%
  left_join(rank_all %>% select(county, rank), by = "county") %>%
  left_join(gini_all %>% filter(year == 2025) %>% select(county, gini), by = "county") %>%
  mutate(
    group = ifelse(county %in% GROUP_A, "Rift Valley", "Central"),
    y2014 = format(round(y2014), big.mark = ","),
    y2025 = format(round(y2025), big.mark = ","),
    gini  = round(gini, 4)
  ) %>%
  arrange(rank) %>%
  select(
    Group  = group,
    County = county,
    `2014 Allocation (KES)` = y2014,
    `2025 Allocation (KES)` = y2025,
    `Growth (%)`      = growth_pct,
    `Nat. Rank (2025)` = rank,
    `Gini (2025)`     = gini)
scorecard %>%
  kbl(align = c("l", "l", "r", "r", "r", "c", "r")) %>%
  kable_styling(full_width = TRUE, bootstrap_options = c("striped", "hover")) %>%
  pack_rows("Rift Valley - Group A", 1, 3) %>%
  pack_rows("Central - Group B", 4, 6)
```

4.5 Discussion of Findings

The findings reveal several important patterns and implications for understanding NGCDF allocations across two regionally distinct county clusters.

Allocation Growth: Both Group A and Group B counties recorded consistent nominal growth in NGCDF allocations over the 2014–2026 period, reflecting Kenya's expanding national budget and the progressive scaling of constituency development funding. However, once the allocations are deflated using Kenya's Consumer Price Index, real growth is considerably more modest

Table 16: All six target counties key metrics scorecard (2025)

Group	County	2014 Allocation (KES)	2025 Allocation (KES)	Growth (%)	Nat. Rank (2025)	Gini (2025)
Rift Valley — Group A						
Central	Kiambu	820,134,807	2,153,440,454	162.6	2	0.0209
Central	Murang'a	596,684,678	1,444,507,734	142.1	8	0.0303
Rift Valley	Uasin Gishu	440,829,780	1,076,651,727	144.2	23	0.0361
Central — Group B						
Rift Valley	Nandi	440,264,275	1,076,651,727	144.5	26	0.0306
Rift Valley	Trans Nzoia	391,921,174	897,209,772	128.9	31	0.0400
Central	Kirinyaga	275,107,712	717,779,318	160.9	35	0.0500

across all six counties, a reminder that headline nominal increases do not always translate into equivalent gains in development purchasing power.

Structural Drivers of County Totals: Absolute county-level totals are heavily influenced by the number of constituencies each county holds. Kiambu, with twelve constituencies, and larger counties in general will structurally out-total counties with fewer constituencies regardless of relative need. This confirms that per-constituency and per-voter measures, rather than absolute totals, are the fairer basis for cross-county equity comparisons, a finding consistent with the Equity Theory discussed in Chapter Two.

Intra-County Equity: Gini coefficients and coefficients of variation reveal that intra-county equity varies both across counties and over time. Counties with a wider mix of urban and rural constituencies, most notably Kiambu, given its proximity to Nairobi — show conditions conducive to higher within-county inequality, while smaller, more agriculturally homogeneous counties tend to show comparatively more even distributions, though this should not be assumed without examining the data for each county and year.

Per-Voter Equity: Per-voter allocation analysis offers a fairer lens than absolute totals for assessing whether registered voters are being served proportionately. Constituencies with smaller voter rolls sometimes receive disproportionately higher per-voter funding even where their absolute allocation is lower, suggesting the equal-share component of the NGCDF formula meaningfully cushions smaller constituencies.

Regional Comparison: The Rift Valley cluster (Group A), being predominantly agricultural and rural, and the Central cluster (Group B), anchored by the more urbanized Kiambu, occupy different positions in Kenya's development geography, yet both receive NGCDF funds through the same national formula. Where the two groups' allocation ranges overlap substantially, this suggests the two regions are broadly within the same tier of national NGCDF receipts;

where one group's counties uniformly exceed the other's, this more likely reflects differences in constituency count than differences in underlying need.

Inflation Sensitivity: The national CPI series used throughout this analysis does not capture county-level price differences. This is a particularly important caveat for Kiambu, where construction costs, land prices, and labour rates are likely to sit above the national average due to its Nairobi-adjacent location, meaning its real allocation may be somewhat overstated relative to what it can actually procure locally. For Murang'a, Kirinyaga, and the Rift Valley counties, the national CPI deflator is likely a closer approximation of the true local cost environment.

Forecasting Implications: Linear projections to 2030 suggest continued nominal growth in all six counties if the 2020–2026 trajectory persists. These projections are indicative planning benchmarks rather than guarantees, since national budget shifts, formula amendments, or economic shocks could cause actual allocations to deviate materially.

Data Quality Implications: The reliance on registered voters, rather than official constituency-level population counts, as a normalizing denominator introduces some uncertainty into the per-voter figures, particularly in rapidly urbanizing constituencies where voter registration may lag or outpace actual population change. This underscores the continued need for improved constituency-level demographic data infrastructure in Kenya.

CHAPTER FIVE: SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents a summary of the key findings from the NGCDF allocation analysis, draws conclusions based on the evidence, and offers recommendations for policy, practice, and future research. The chapter also acknowledges limitations of the study and suggests directions for further investigation.

5.2 Summary of Findings

The study analyzed NGCDF allocations across the six study counties over the period 2014–2026, with linear projections through 2030, yielding several important findings aligned with the research objectives.

Objective 1: Document allocation totals and year-on-year trends. All six counties recorded consistent nominal growth in NGCDF allocations across the study period, with growth trajectories fitted using OLS regression on the 2020–2026 sub-period to generate 2027–2030 projections. Kiambu recorded the highest absolute totals within Group B, driven substantially by its twelve constituencies.

Objective 2: Analyze constituency-level distribution and performance tiers. Constituency-level tier analysis classified constituencies within each county into High, Mid, and Low relative performance tiers based on their 2025 allocation, revealing meaningful intra-county variation that absolute county totals alone would obscure.

Objective 3: Assess intra-county equity and nominal vs. real growth. Gini coefficients and coefficients of variation revealed that intra-county equity varies both across counties and over time, with no single county showing uniformly high or low inequality across all years. CPI-adjustment of allocations confirmed that real (inflation-adjusted) growth has been considerably more modest than nominal growth across all six counties, given Kenya's cumulative inflation of approximately 77% between 2014 and 2025.

Objective 4: Provide a comparative assessment between Group A and Group B. Cross-group comparison revealed that structural factors — chiefly the number of constituencies per county — are a major driver of absolute allocation differences between the Rift Valley and Central clusters. Per-voter and national ranking analysis provided a fairer basis for assessing whether the two regional clusters are being served equitably by the national NGCDF formula.

5.3 Conclusions

Based on the findings, several conclusions can be drawn regarding NGCDF allocations in the six study counties. Overall, allocations increased steadily between 2014 and 2026 in nominal terms. However, after adjusting for inflation, real growth was much smaller, indicating that much of the increase reflects rising prices rather than a substantial expansion in funding.

The analysis also shows that equity in allocation varies across counties. Gini coefficients and coefficients of variation reveal differences in how fairly funds are distributed among constituencies, making these useful indicators for monitoring disparities over time. In addition, per-voter allocations provide a more meaningful measure of equity than absolute totals, as they account for differences in constituency voter populations.

National rankings place the study counties within the broader context of all 47 Kenyan counties, while forecast results suggest that nominal allocations are likely to continue increasing through 2030, subject to national budget priorities and economic conditions. Finally, the comparison between Rift Valley and Central counties indicates that the number of constituencies is a stronger determinant of total NGCDF allocations than regional location, highlighting the importance of using per-constituency and per-voter measures when assessing allocation equity.

5.4 Recommendations

5.4.1 Policy Recommendations

Allocation Formula Review: The NGCDF Board should periodically review the allocation formula to ensure it adequately addresses equity objectives across counties with differing constituency counts and urbanization levels. The review should consider whether the current balance between the equal-share, population, and poverty components adequately compensates smaller counties, and whether the formula should more explicitly account for local cost-of-living differences, particularly for Nairobi-adjacent counties such as Kiambu.

Inflation-Adjusted Reporting: The NGCDF Board and related agencies should routinely publish CPI-adjusted allocation figures alongside nominal figures, so that stakeholders can distinguish genuine real-terms growth from increases that merely track inflation.

Equity Audits: Regular equity audits, using metrics such as the Gini coefficient and coefficient of variation, should be conducted at the county level to assess whether intra-county distribution is becoming more or less equitable over time, and to flag counties or constituencies warranting closer scrutiny.

Data Infrastructure Improvement: Government agencies should collaborate to improve the availability of constituency-level population data, reducing reliance on registered-voter counts as a population proxy for per-capita analysis.

5.4.2 Practical Recommendations

For NGCDF Management: Implement standardized, machine-readable data reporting across all constituencies, and develop internal capacity for ongoing trend, equity, and forecasting analysis using the framework developed in this study.

For Civil Society Organizations: Utilize the analytical framework developed in this study — combining trend analysis, tiering, Gini/CV equity tracking, CPI adjustment, and forecasting — for independent monitoring of NGCDF allocations in other counties.

For Development Partners: Support efforts to improve constituency-level demographic data infrastructure and fund research extending this analytical framework to additional county clusters.

For Researchers: Extend the analysis to all 47 counties, incorporate development outcome data to assess whether allocations translate into improved conditions, and investigate the political economy factors (e.g., electoral cycles) that may influence allocation patterns.

5.5 Limitations of the Study

Several limitations should be considered when interpreting the findings:

Population Data Constraints: The absence of routinely available constituency-level population data required the use of registered-voter counts as a normalizing proxy, which may not perfectly reflect actual population, particularly in rapidly urbanizing constituencies.

National-Level CPI: The CPI index used is a national Kenya-wide measure and does not capture county-level price differences, which are particularly relevant for Kiambu given its proximity to Nairobi.

Forecast Uncertainty: Linear projections for 2027–2030 are indicative rather than predictive; formal confidence intervals were not reported given the exploratory nature of the projections and the short (2020–2026) estimation window.

Allocation Data Scope: The analysis examined allocation data only, not actual disbursements or expenditure patterns, which may differ from allocations due to disbursement delays or other implementation factors.

Outcome Assessment: The study did not assess the impact or effectiveness of NGCDF-funded projects on development outcomes, limiting conclusions about whether allocations translate into improved conditions on the ground.

Political Economy Factors: The analysis did not examine political factors — such as the proximity of elections or the political alignment of constituency representatives — that may influence allocation patterns.

5.6 Suggestions for Further Research

Building on this study, several directions for future research emerge:

National Extension: Extend the analysis to all 47 counties, enabling a comprehensive, nationwide assessment of NGCDF allocation patterns and equity.

Outcome Evaluation: Examine the relationship between allocations and development outcomes (e.g., school enrolment, health facility access, water access) to assess whether resources translate into improved conditions.

Political Economy Analysis: Investigate the role of electoral cycles, Member of Parliament tenure, and political alignment in shaping allocation patterns across counties.

Sub-National CPI: Develop or apply county-level or regional cost-of-living indices to refine the real-allocation analysis, particularly for rapidly urbanizing counties such as Kiambu.

Disbursement Analysis: Track actual disbursements versus allocations to examine factors affecting fund absorption and project implementation.

Comparative Fund Analysis: Compare NGCDF allocations with other development funding streams available to the same counties, including county government development budgets and sector-specific national funds.

By pursuing these research directions, scholars and practitioners can continue to build understanding of NGCDF dynamics and contribute to improved development outcomes through more effective and equitable resource allocation in Kenya's diverse regions.

REFERENCES

Bagaka, O. (2009). *Fiscal decentralization in kenya and the growth of government: The constituency development fund* [PhD thesis]. Northern Illinois University.

Cheeseman, N., Kanyinga, K., & Lynch, G. (2020). The political economy of kenya: Community, clientelism, and class. In N. Cheeseman, K. Kanyinga, & G. Lynch (Eds.), *The oxford handbook of kenyan politics*. Oxford University Press.

Commission on Revenue Allocation. (2022). *Recommendation on the basis for equitable sharing of revenue between the national and county governments*. Commission on Revenue Allocation.

Independent Electoral and Boundaries Commission. (2022). *Register of registered voters by constituency*. IEBC.

Kenya National Bureau of Statistics. (2020). *Economic survey and statistical abstract*. KNBS.

Kenya National Bureau of Statistics. (2025). *Consumer price indices and inflation rates*. KNBS.

Kimenyi, M. S. (2005). *Efficiency and efficacy of kenya's constituency development fund: Theory and evidence* (Economics Working Paper 2005-42). University of Connecticut, Department of Economics.

Musgrave, R. A. (1959). *The theory of public finance: A study in public economy*. McGraw-Hill.

Mwangi, [FIRST. N., & Otieno, [FIRST. N. (2022). [UNVERIFIED — supply exact title of the NGCDF allocation study you cited]. *[UNVERIFIED — Journal Name]*.

National Government Constituencies Development Fund Board. (2025). *Constituency allocation data, financial years 2014–2026*. NGCDF Board.

Oates, W. E. (1972). *Fiscal federalism*. Harcourt Brace Jovanovich.

Odhiambo, [FIRST. N. (2021). [UNVERIFIED — supply exact title of the NGCDF governance study you cited]. *[UNVERIFIED — Journal Name]*.

Rawls, J. (1971). *A theory of justice*. Harvard University Press.

Republic of Kenya. (2015). *National government constituencies development fund act, no. 30 of 2015 (cap. 414A)*. Kenya Gazette Supplement No. 200; Government Printer.

Sen, A. (1992). *Inequality reexamined*. Harvard University Press.

Smoke, P. (2001). *Fiscal decentralization in developing countries: A review of current concepts and practice* (Democracy, Governance and Human Rights Programme Paper 2). United Nations Research Institute for Social Development (UNRISD).

Tiebout, C. M. (1956). A pure theory of local expenditures. *Journal of Political Economy*, *64*(5), 416–424. https://doi.org/10.1086/257839

United States Agency for International Development. (2021). *Assessment of devolution and local governance in kenya*. USAID.

World Bank. (2020). *Kenya public expenditure review: Options for fiscal consolidation*. World Bank.

APPENDICES

Appendix I: Data Dictionary

Variable Name	Description	Source
county	County name	IEBC
constituency	Constituency name	IEBC
voters	Registered voters	IEBC
2014 … 2026	NGCDF Allocation by financial year (KES)	NGCDF Board
year	Financial year (long format)	NGCDF Board
alloc	NGCDF Allocation (KES, long format)	NGCDF Board
cpi	Consumer Price Index (2014 = 100)	KNBS
real_total	CPI-deflated (real, 2014 KES) county total allocation	Derived
per_voter	Allocation per registered voter (KES)	Derived
gini	Gini coefficient of constituency allocations	Derived
cv	Coefficient of Variation (%) of constituency allocations	Derived
tier	Constituency performance tier (High / Mid / Low)	Derived
rank	National rank by total 2025 allocation (all 47 counties)	Derived

Appendix II: Group and County Codes

Group	County	Constituencies
Group A — Rift Valley	Trans Nzoia	5
Group A — Rift Valley	Nandi	6
Group A — Rift Valley	Uasin Gishu	6
Group B — Central	Murang'a	10
Group B — Central	Kiambu	12
Group B — Central	Kirinyaga	7

Appendix III: R Session Information

```
R version 4.5.2 (2025-10-31 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)

Matrix products: default
  LAPACK version 3.12.1

locale:
[1] LC_COLLATE=English_United States.utf8
[2] LC_CTYPE=English_United States.utf8
[3] LC_MONETARY=English_United States.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.utf8

time zone: America/Halifax
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
 [1] here_1.0.2       janitor_2.2.1    kableExtra_1.4.0 knitr_1.50
 [5] ineq_0.2-13      patchwork_1.3.2  scales_1.4.0     lubridate_1.9.4
 [9] forcats_1.0.1    stringr_1.6.0    dplyr_1.2.1      purrr_1.2.2
[13] readr_2.1.6      tidyr_1.3.2      tibble_3.3.0     ggplot2_4.0.3
[17] tidyverse_2.0.0

loaded via a namespace (and not attached):
 [1] generics_0.1.4     xml2_1.4.1         lattice_0.22-7     stringi_1.8.7
 [5] hms_1.1.4          digest_0.6.39      magrittr_2.0.4     evaluate_1.0.5
 [9] grid_4.5.2         timechange_0.3.0   RColorBrewer_1.1-3 fastmap_1.2.0
[13] Matrix_1.7-4       rprojroot_2.1.1    jsonlite_2.0.0     mgcv_1.9-3
[17] viridisLite_0.4.2  textshaping_1.0.4  cli_3.6.6          crayon_1.5.3
[21] rlang_1.2.0        splines_4.5.2      bit64_4.6.0-1      withr_3.0.2
[25] yaml_2.3.10        parallel_4.5.2     tools_4.5.2        tzdb_0.5.0
[29] vctrs_0.7.3        R6_2.6.1           lifecycle_1.0.5    snakecase_0.11.1
[33] bit_4.6.0          vroom_1.6.6        pkgconfig_2.0.3    pillar_1.11.1
[37] gtable_0.3.6       glue_1.8.0         systemfonts_1.3.1  xfun_0.54
[41] tidyselect_1.2.1   rstudioapi_0.18.0  farver_2.1.2       nlme_3.1-168
[45] htmltools_0.5.8.1  labeling_0.4.3     rmarkdown_2.30     svglite_2.2.2
```

```
[49] compiler_4.5.2     S7_0.2.1
```